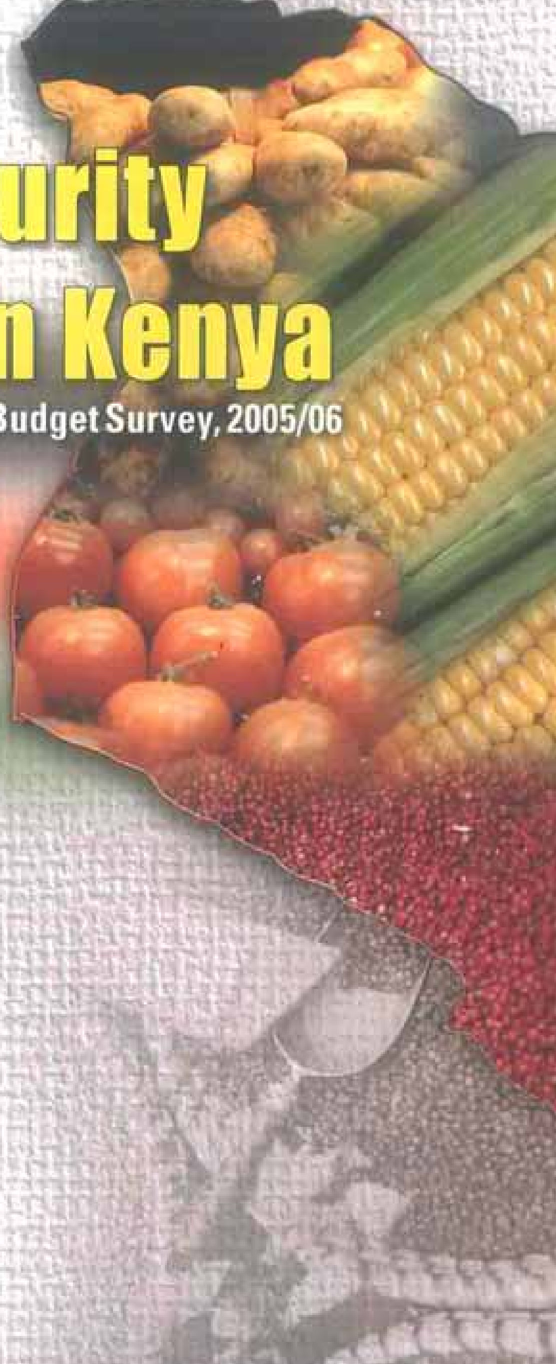
REPUBLIC OF KENYA

Kenya National Bureau of Statistics

Food Insecurity Assessment in Kenya

Based on Kenya Intergrated Household Budget Survey, 2005/06



April 2008, Nairobi

The EC-FAO Food Security "Information for Action" Programme is funded by the European Union and implemented by FAO

Republic of Kenya

Kenya National Bureau of Statistics
Nairobi, Kenya

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FOREWORD

Ten years ago, world leaders met in Rome for the World Food Summit (WFS) to discuss ways to end hunger. They pledged their commitment to an ongoing effort to eradicate hunger in all countries and set themselves the target of halving the number of undernourished people by 2015. Countries also committed themselves to the Millennium Development Goal (MDG 1) of halving hunger and extreme poverty by the year 2015.

This publication presents information about the country's food insecurity situation at national and sub-national levels using FAO methodology. The information is based on food consumption statistics derived from data collected in the Kenya Integrated Household Budget Survey (KIHBS), 2005/06. The trends presented in this report show the extent of the country's food insecurity in relation to the FAO/WHO food requirements. It gives at a glance the country's food deprivation, which is the measure of prevalence of undernourishment using FAO methodology.

I wish to congratulate the FAO office in Rome for its support towards the analysis and tabulation of data used in this report. In addition, I wish to thank Ministry of Agriculture, Ministry of Livestock and Fisheries Development, Ministry of Health for their collaboration with the Kenya National Bureau of Statistics in preparing this report.

Edward Sambili, CBS
Permanent Secretary
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ACKNOWLEDGEMENT

This report provides information about the Country's food insecurity situation at the national and sub-national levels, using Food and Agriculture Organization methodology based on food consumption data collected in the 2005/06 Kenya Integrated Household Budget Survey (KIHBS 2005/06).

The Bureau is greatly indebted to the Food and Agriculture Organization (FAO) for supporting the process of producing this report and in particular, Mr. Seevalingum Ramasawmy and Ms Nathalie Troubat of FAO, Rome for providing technical assistance.

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ACRONYMS

ADEC	Average Dietary Energy Consumption
CPF	Critical Food Poverty
CV	Coefficient of variation
DEC	Dietary Energy Consumption
EC	European Community
FAO	Food and Agriculture Organization
FBS	Food Balance Sheet
FMV	Food Monetary Value
FSSM	Food Security Statistical Module
KIHBS	Kenya Integrated Household Budget Survey
KNBS	Kenya National Bureau of Statistics
KSh	Kenya Shilling
MDER	Minimum Dietary Energy Requirement
MDG	Millennium Development Goals
UNU	United Nations University
USDA	United States of America Department of Agriculture
WFS	World Food Summit
WHO	World Health Organization

EXECUTIVE SUMMARY

Although the Country's economy has improved significantly during the last five years, food insecurity is still quite widespread among its population. In 2005, the estimate of the overall food deprivation, which is the measure of MDG Indicator 5 on prevalence of under nourishment, was 51 per cent. Food deprivation in rural areas was 57 per cent, much higher than 39 per cent in the urban areas.

Food dietary energy consumption at national level was 1800 kcal/person/day. The urban population had higher dietary energy consumption of 2060 kcal/person/day than the rural population's, 1690 kcal/person/day.

The minimum dietary energy requirement (MDER) at national level was 1683 kcal/person/day, slightly higher than that of the rural population 1670 kcal/person/day, but lower than that of the urban population's, 1733 kcal/person/day.

The prevalence of critical food poverty (CFP), which is a measure of food income deprivation, was 24 per cent at the national level (proportion of population with insufficient income to acquire a balanced food basket yielding the MDER). The prevalence of critical food poverty for the rural population was 22 per cent while that of urban population was 15 per cent.

The dietary energy consumption of the food insecure population was 1261 kcal/person/day while the depth of hunger, which is the shortfall to the MDER, 1683 kcal/ person/day, was 422 kcal / person / day.

The share of food monetary value to total consumption was 46 per cent at the national level. This share for the lower income quintile was 76 per cent, higher than the upper quintile, 30 per cent.

The country acquired 15 per cent of their food consumption from its own production. This pattern was far higher in rural areas, at 23 per cent, compared to 3 per cent in the urban areas. Food purchased accounted for 72 per cent of the total food acquisition in urban areas.

Nationally, 67 per cent of dietary energy was derived from carbohydrates, 12 per cent from proteins and 21 per cent from fats, which is within the international guidelines by WHO/FAO. The national average consumption levels of carbohydrates, proteins and fats were 305, 53 and 42 grams/person/day respectively. Cereals and cereal products contributed about 48 per cent of the total dietary energy consumption.

The inequality in access to food due to income measured as the coefficient of variation of dietary energy consumption (CV) at national level was 35 per cent and the Gini coefficient of food dietary energy consumption due to income was 20 per cent.

CHAPTER 1

Introduction

The Kenyan economy has relatively advanced agricultural and industrial sectors. It also derives substantial foreign exchange earnings from agricultural exports and tourism. After decades of low economic growth, Kenya has seen economic recovery since 2003 and registered about 6.1 per cent growth in Gross Domestic Product (GDP) in 2006. The main agricultural products are tea, coffee, sugarcane, horticultural products, wheat, rice, sisal, pineapples, pyrethrum, dairy products, meat and meat products, hides, skins which account for about 25 per cent of the national GDP. A large proportion of Kenya's land is arid or semi-arid and is the home of pastoral and nomadic people. The country is vulnerable to unstable rain patterns which affect greatly food production. Subsistence farming is very high in rural Kenya and probably the primary source of livelihood to the population in rural areas.

More than half of the country's 37 million people live in rural areas where there are a high proportion of poor people with low standards of living and limited access to social services. Almost 75 per cent of the Kenyan labour force is in the agriculture sector. Poverty remains an enormous challenge for Kenya, though measures are being put in place to correct the situation.

In 2005/06 the Kenya National Bureau of Statistics conducted the Kenya Integrated Household Budget Survey (KIHBS) that collected data on food consumption among others. The data were used to derive a wide range of indicators on food insecurity at national and sub national levels to identify food insecure people and assess their level of vulnerability. Data were processed and analysed using the Food Security Statistics Module software that was developed by the FAO statistics Division following the FAO methodology on estimation of food deprivation. This report presents the main findings.

CHAPTER 2

SURVEY METHODOLOGY

2.1 The Kenya Integrated Household Budget Survey

The Kenya National Bureau of Statistics has been conducting Welfare Monitoring Survey (WMS) series with WMS I in 1992, WMSII in 1994 and WMS III in 1997. The 2005/06 Kenya Integrated Households Budget Survey (KIHBS) included data on income, expenditure, employment, health, education, agriculture production, social amenities (water and sanitation) and food consumption among others. The coverage was countrywide and covered 12 months, from May, 2005 to May, 2006 and hence accounted for all the seasons. The sample size was 13,430 households. The survey was conducted in 17 cycles with each cycle covering 21 days. Sampled households for the survey, were interviewed once within each cycle.

2.2 Food Consumption Data

The KIHBS collected a wide range of household consumption expenditures including food consumption. The questionnaire contained the 7-day recall on food data from purchase, own production, gift and meals eaten out. It also distinguished out of what was purchased how much was consumed in the household in the reference 7 days. The 21 day cycle contained 14 days of consumption and purchases diaries completion and filling of the main household questionnaire. Data capture and editing was done in the field using laptops thus making verification possible in the field.

The 2005/2006 KIHBS data collection was based on two distinct approaches namely, food acquisition and food consumption. A seven day recall questionnaire collected the monetary and quantity values of food purchases during a seven-day-period. The questionnaire also collected the quantity of consumed food obtained from purchases, own production, own purchased stocks, as well as gifts and meals eaten out. Food details were available for a list of 154 food items which are common in Kenyan households including food items consumed away from home.

Food consumption data instead of food purchases data was used for the purposes of analysis in this report. This is because part of food purchased is used for other purposes in the household rather than consumption. The KIHBS (2005/06) survey collected information on food consumption some of whose quantities were in non standard unit of measure and all had to be converted for standardization. The non standard unit conversion factors were obtained from the market questionnaire which was administered in every cluster. Using market prices collected during the survey all the food sources in the following categories; own consumption, gifts and own stock were calculated for their monetary values. The Kenya Composition Table and the African Food composition Tables were used to convert food quantities into nutrient values. Additionally in this analysis, the total household expenditure is assumed to be equivalent to income. The resulting food consumption data was processed and analysed using the FAO-Food Security Statistics Module (FSSM) to derive the food security indicators at national and sub-national level.

2.3 Sample design and survey coverage

A total of 13,430 households were randomly selected to comprise the KIHBS sample, which was designed to generate representative statistics at the national and sub national levels. The total sample sizes in rural and urban areas were 8,610 and 4,820 households respectively. Upon completion of fieldwork, two adjustments were made to the sampling weights. Firstly, some of the sampled households did not participate in the Survey, either because of failure to establish contact or explicit refusal to participate. This is a common feature in all household surveys and is called "unit non-response". The KIHBS anticipated this would happen and during the sampling design phase, five replacement households were randomly selected in each KIHBS cluster. These were used as substitutes for original households that did not respond or were not available. Overall the effective coverage and response rate was about 98 per cent. However, fourty of the sampled 13,430 households could not be interviewed because of either insecurity or inaccessibility of the clusters.

CHAPTER 3

Results and Findings

3.1. Magnitude of food deprivation

Food insecure people are those individuals whose food intake falls below their minimum dietary energy requirements.

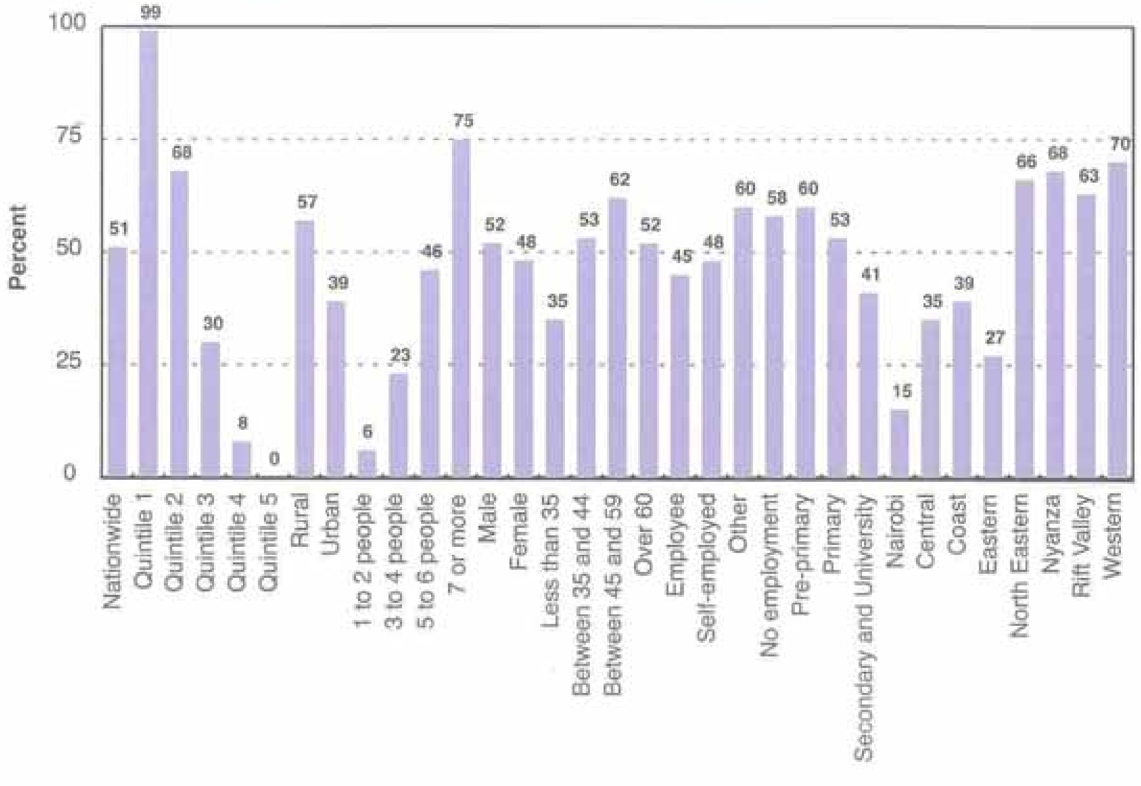
Millennium Development Goal 1 on hunger targets to halve, between 1990 and 2015, the proportion of people who suffer from hunger and is specified as Indicator 5:

"Proportion of population below minimum level of dietary energy consumption"

The Minimum Dietary Energy Requirement (MDER) is the amount of dietary energy of an average individual that is considered adequate to meet his daily energy needs for light activity and good health given his/her age and sex.

The magnitude of hunger (food deprivation) as measured by the prevalence of undernourishment shows that in 2005/06 about one person out of two was undernourished in Kenya. About 51 per cent of the population had food consumption below the Minimum Dietary Energy Requirement (MDER). The daily MDER for the nation was estimated on the minimum dietary energy needed to maintain body-weight and perform a sedentary light physical activity taking account the age and sex structure of the population. The MDER was 1683 kcal/person/day at national level according to 2005/06 KIHBS. Levels of food deprivation decreased with rises in income levels, with food deprivation of 99 per cent for the lowest income group to less than one per cent of undernourished people in the highest income group as shown in Figure 1.

Figure 1: Proportion of food deprivation in total population (2005/06)



Urban areas had a significantly lower prevalence of undernourishment of 39 per cent compared to 57 per cent in rural areas where majority of the population live. Large households having seven or more members had a high level of undernourishment of 75 per cent. The households whose heads were economically inactive showed higher prevalence of undernourishment than those with economically active heads. For households whose heads were employed, 45 per cent were undernourished compared to 58 per cent for non-employed.

Analysis of food deprivation by gender showed that male headed households had 52 per cent prevalence of undernourishment compared to 48 per cent for female headed households. Households whose heads had secondary or higher education showed lower undernourishment levels than those with primary or special/incomplete higher education. At the regional level, Nairobi recorded the least undernourishment level of 15 per cent, whereas Western province registered the highest at 70 per cent.

3.2 Depth of hunger

In 2005/06, 51 per cent of the population recorded a daily food consumption of 1261 kcal/person, which was well below the national MDER benchmark of 1683 kcal/person/day. This implied that they required an additional 422 kcal/day to reach the national MDER and another 117 kcal/day to get to the national average dietary energy consumption of 1800 kcal/person/day as shown in Figure 2.

Figure 2: Depth of Hunger (2005/06 KIHBS)

Figure 3 below shows the gap between the Average Dietary Energy consumption of undernourished people from urban and rural areas with respective MDER. The MDER of urban populations was 1733 kcal/person/day compared to 1670 kcal/person/day for rural populations. The average DEC of food deprived people in urban areas was 1356 kcal/person/day and 1224 kcal/person/day in rural areas, which meant urban dwellers required 377 kcal/person/day to reach the MDER, while rural needed 446 kcal/person/day.

Figure 3: Depth of hunger in urban and rural areas of Kenya

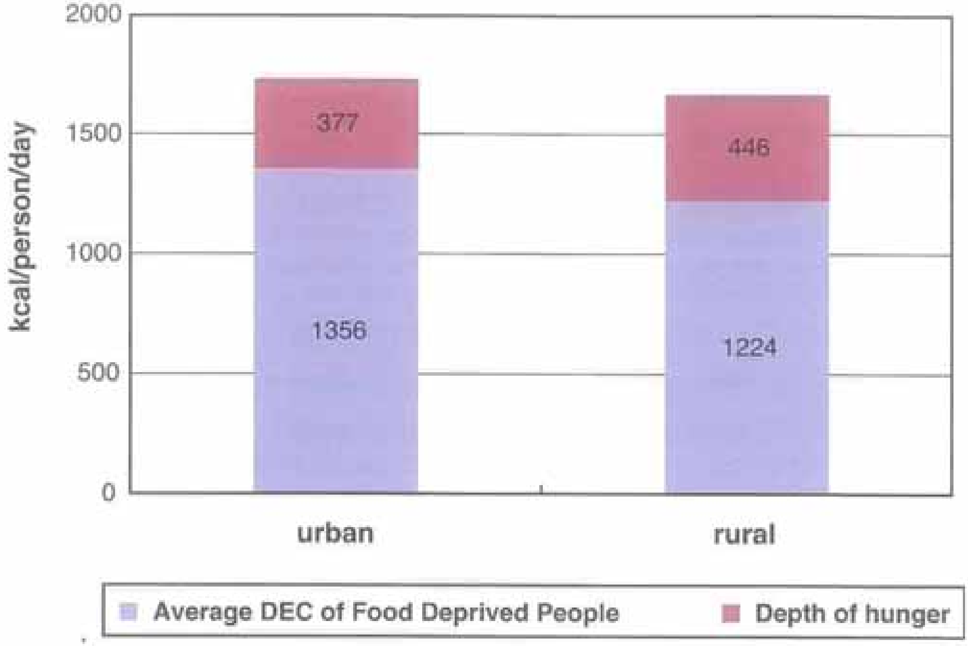


Figure 4a: ADEC and MDER by regions (2005/06 KIHBS)

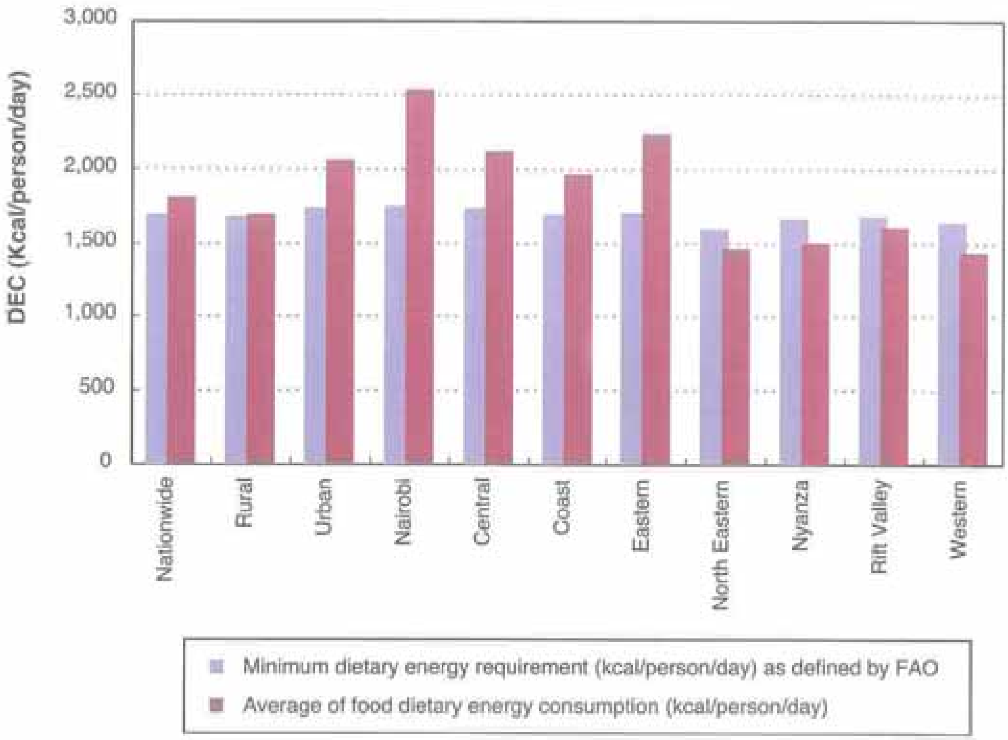


Figure 4a compares the average dietary energy consumption (ADEC) to the minimum dietary energy requirement (MDER) by regions.

The Minimum Dietary Energy Requirements ranged from 1745 kcal/person/day in Nairobi to 1595 kcal/person/day in North Eastern, a difference of 150 kcal/person/day. However, the Average Dietary

Energy Consumption showed a significant variation of 1090 kcal with Nairobi registering the highest at 2530 kcal/person/day and the lowest in Western at 1440 Kcal/person/day. Rift valley, Nyanza, North Eastern and Western provinces recorded ADEC below national MDER. These variations are indicative of problems of access and distribution of food consumption.

Figure 4b shows the disparities among demographic and socio-economic sub national population groupings. The range of variation in the average DEC of the food insecure population was of smaller magnitude than among the geographical regions. This implies that there are regional hunger spots in the Country requiring food interventions. Male headed households in urban areas recorded the lowest average DEC. Large households having more than four members had an average DEC almost equal to their needs, but its food insecure population having a relatively high average DEC. These households did not have sufficient food due to inadequate income.

3.3. Critical Food Poverty

The prevalence of Critical Food Poverty (pCFP) is the proportion of the population whose income is lower than the cost of a macronutrient-balanced food basket equivalent to the minimum dietary energy requirement. The MDER cost is valued using macro-nutrient unit costs from food consumed by households in the first income quintile. The macronutrient-balanced food basket provides 12.5, 22.5 and 65 per cent energy from proteins, fats and carbohydrates respectively. The monetary value of 1000 kcal corresponding to a balanced diet for the population in the first quintile equalled KSh 16.07 raising the poverty line at about KSh 27.04 meaning that about 24 per cent of the population did not have enough money to access a food basket corresponding to minimum requirements in terms of energy and nutrients. The proportion of people not having sufficient income to acquire a balanced diet basket of food equivalent to the minimum dietary energy requirement is based on the income distribution. The different levels of critical food poverty estimated using the probability framework of the income distribution for sub national groups of the population and at regional level are shown in figure 5.

Figure 4b: MDER and ADEC by Sub-national Groupings (2005/06 KIHBS)

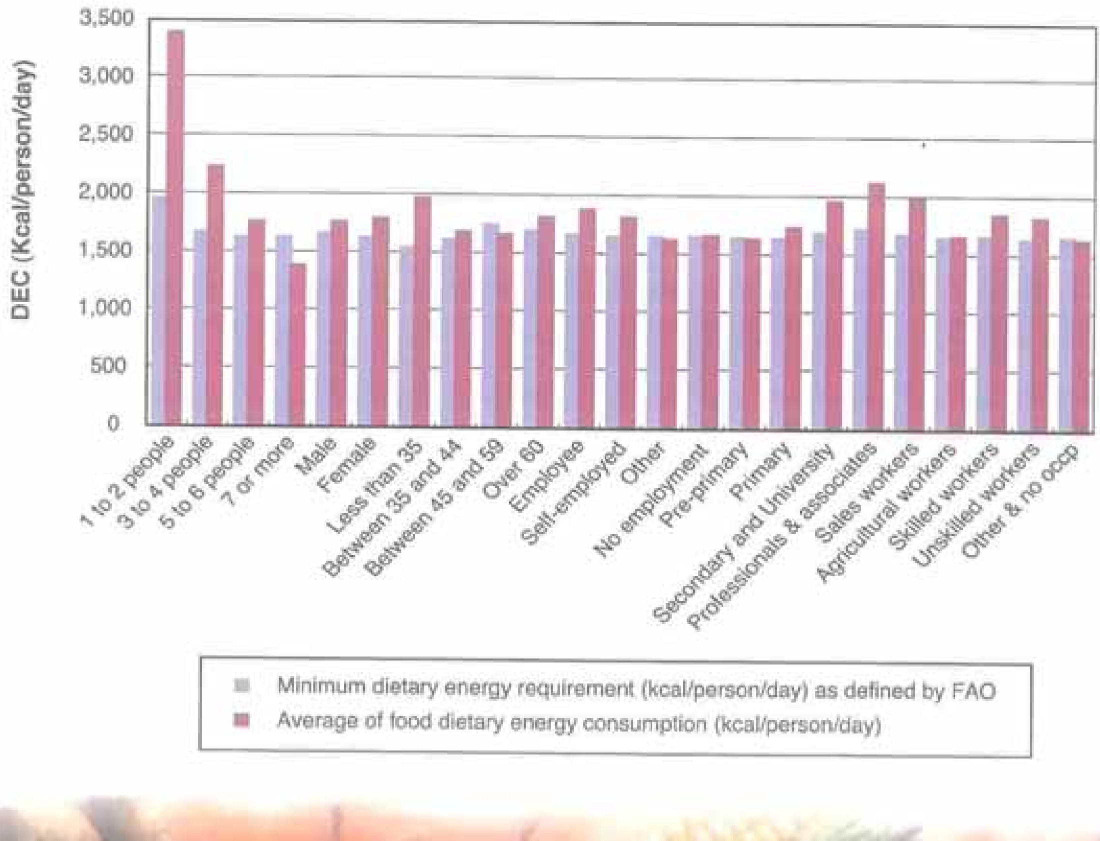
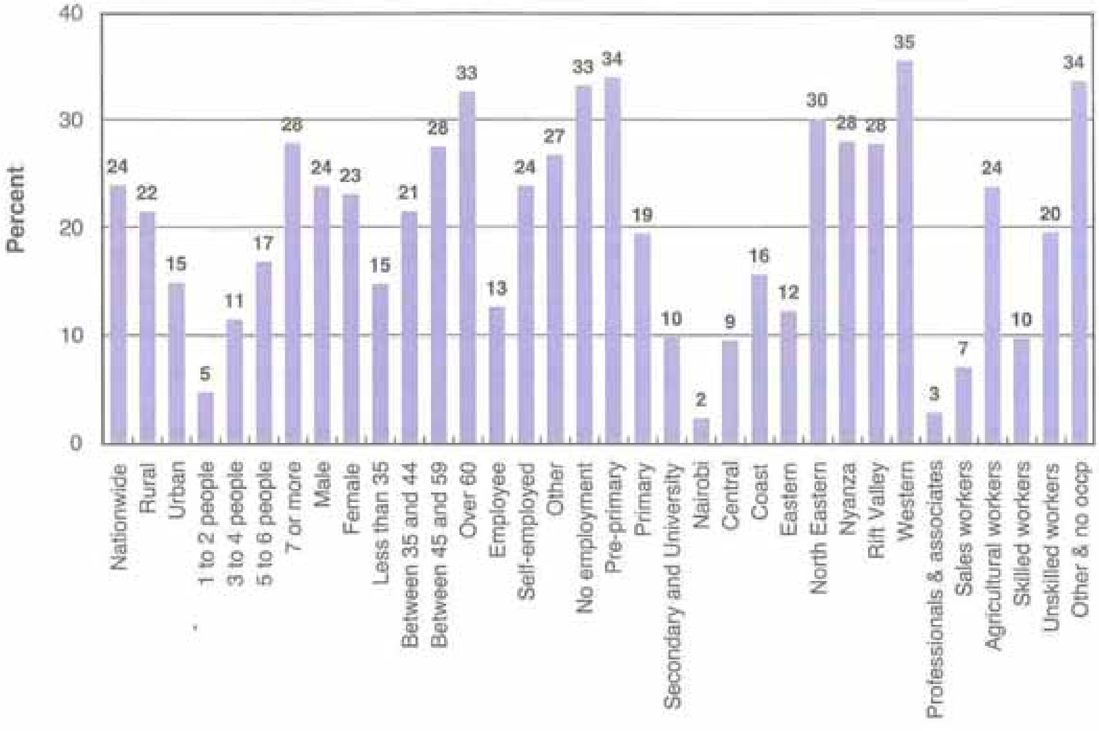


Figure 5: Critical food poverty by Sub-national Groupings



The level of critical food poverty in urban areas was lower than that in rural areas with respectively 15 and 22 per cent. It was among households of large size whose head had low education or was over sixty or was unemployed that the levels of critical food poverty were the highest at about 30 per cent. Levels of critical food poverty differed also according to regions as the lowest level of critical food poverty was observed in Nairobi at 2 per cent while the Western province recorded the highest level at 35 per cent.

3.4. Food consumption and expenditure

3.4.1. Dietary Energy Consumption (DEC)

According to 2005/06 KIHBS as shown in Figure 6, the daily average Dietary Energy Consumption (DEC) was 1800 kcal/person. The dietary energy consumption level in rural and urban areas was 1690 and 2060 kcal/person/day respectively. Analysis by income levels indicated that DEC increased with rising income levels ranging from 1080 kcal/person/day for people in the lowest income quintile to 3020 kcal/person/day for those in the highest income quintile. Average daily DEC for female headed households was slightly higher than that of male headed households by 30kcal/person.

The average DEC for households' whose heads' were employed, self-employed and not employed were 1910 kcal/person/day, 1840 kcal/person/day and 1690 kcal/person/day respectively. This could be attributed to stable income by the households' whose heads' were employed, hence improved food accessibility.

Consumption by households in the first four income level deciles as shown in Figure 7 were respectively 765, 427, 204 and 27kcal/person/day, lower than the Minimum Dietary Energy Requirement of 1683 kcal/person/day. The four lower income deciles constituted the proportion of households that were food insecure. Households of the ninth and tenth deciles who could afford high energy food commodities were consuming 1086 and 1647 kcal/person/day above the MDER.

Figure 6: Average Dietary Energy Consumption

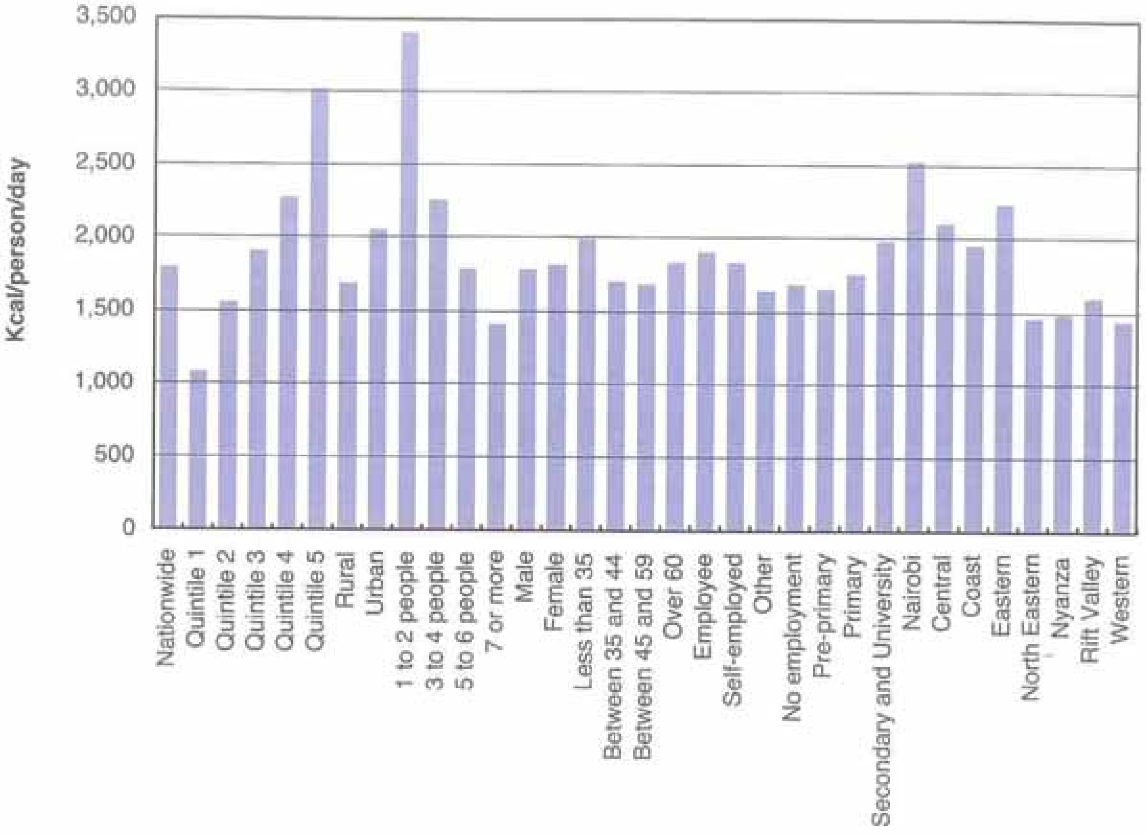
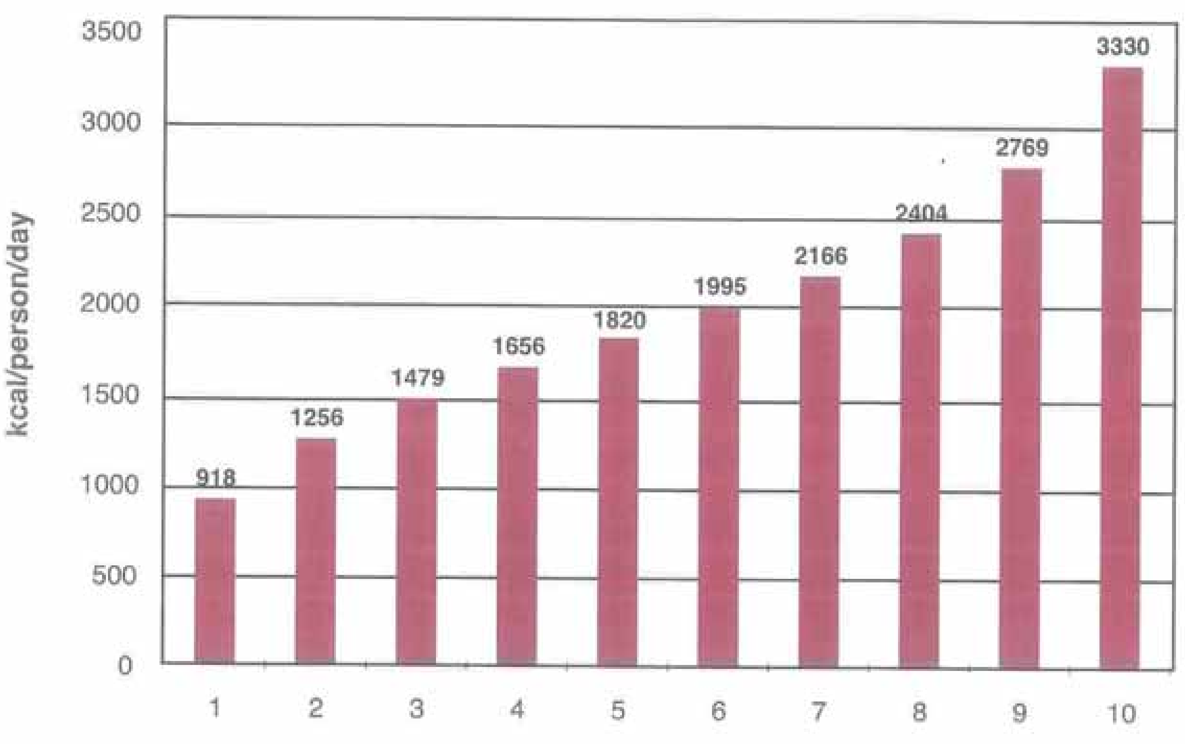


Figure 7: Average dietary energy consumption by deciles of income level

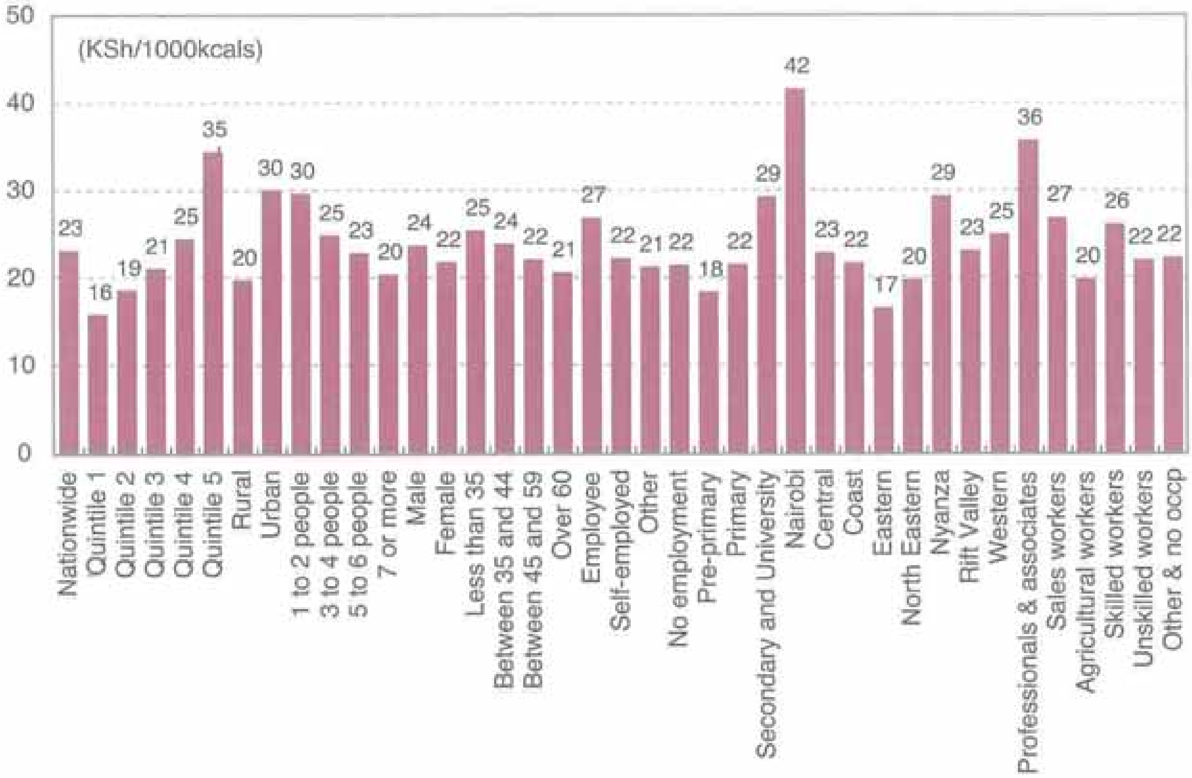


3.4.2. Dietary energy unit value

The average cost of acquiring 1000 kcal in 2005/06 in Kenya as shown in Figure 8 was KSh 23.20. The dietary energy unit value excluded preparation costs such as cost of fuel and labour as they varied among the different income quintiles. With rising income, households consumed food items of better quality and more expensive than those consumed by low income households. Households in the highest income quintile spent on average KSh 34.50 to acquire 1000 kcal while those on the lowest quintile spent KSh 15.90.

The average dietary unit values in rural and urban areas were KSh 19.80 and KSh 30.10 respectively. This is consistent with the average DEC. The high dietary unit cost in urban areas was probably due to market expenses and transport costs for food produced from rural regions, particularly vegetables, fruits and livestock products.

Figure 8: Dietary energy unit monetary value of food



3.4.3. Monetary value of food consumed and Engel ratio

Figure 9 shows monetary value of food consumption. The daily cost of a food basket corresponding to the average dietary energy consumption was KSh 41.70. Significant differences were observed between the lowest income quintile expenditure of KSh 17.20 for an average DEC of 1080 kcal and the highest income group expenditure of KSh 104.30 for an average DEC of 3020 kcal. There was a difference of KSh 28.70 of food consumed per person between urban and rural regions, while a difference of KSh 2.60 was observed between the male and female headed households. Households with small size spent more in acquiring food consumed by each person. The households whose heads had higher education and those with low education spent KSh 58.30 and KSh 30.60 on food respectively.

Figure 9: Monetary value of food consumption

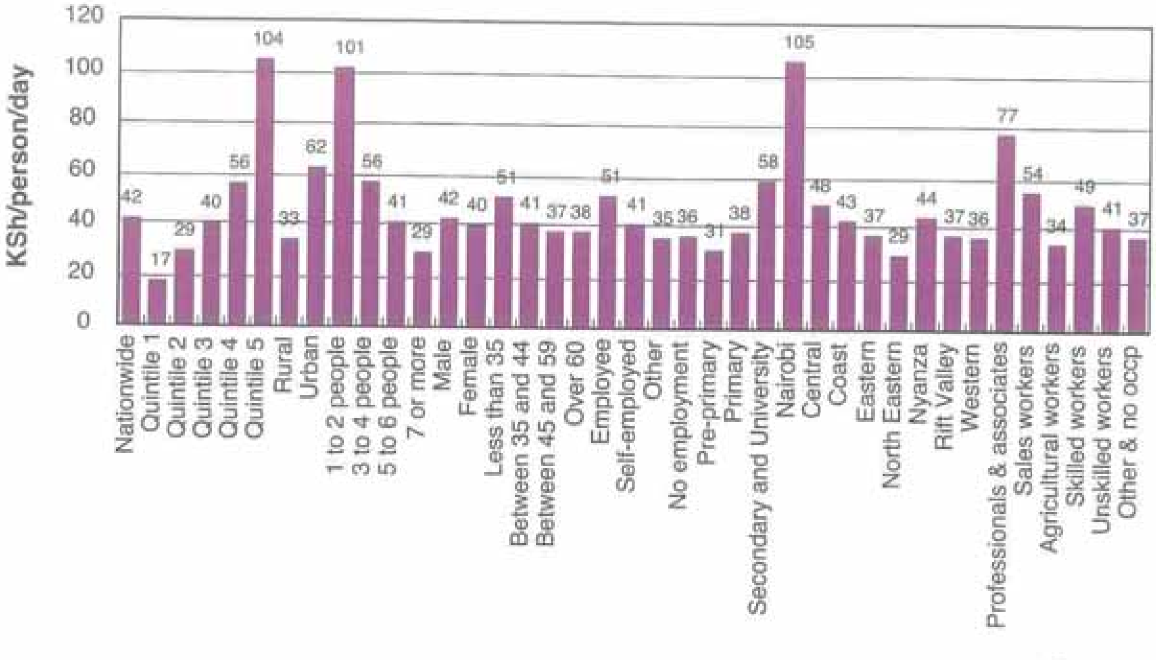
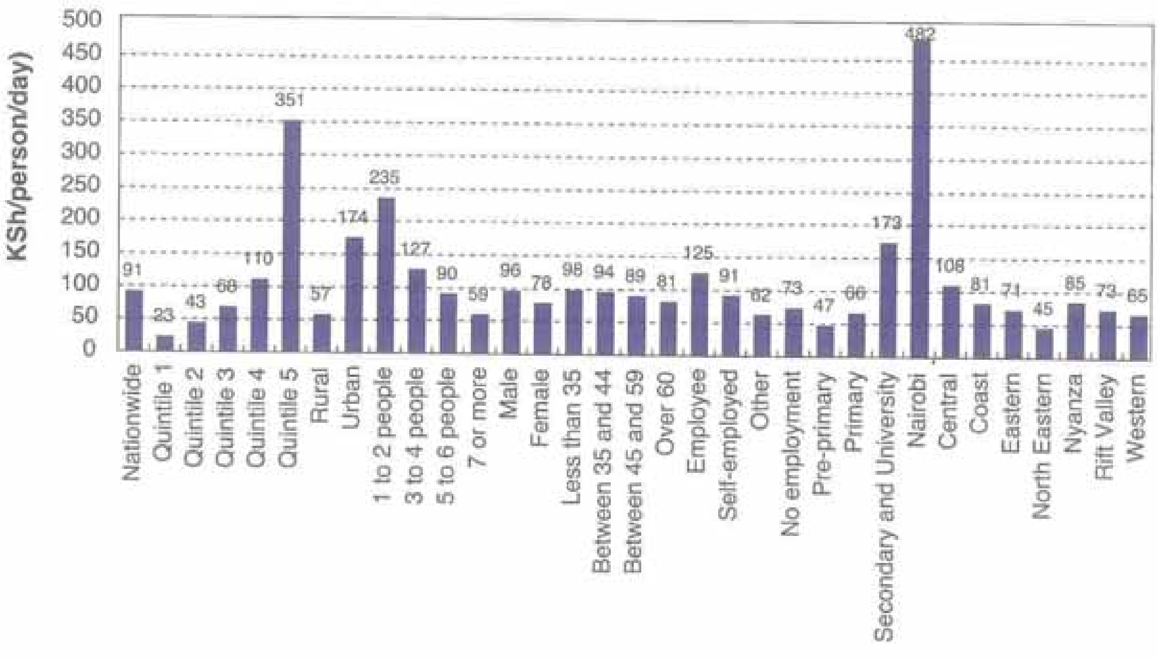


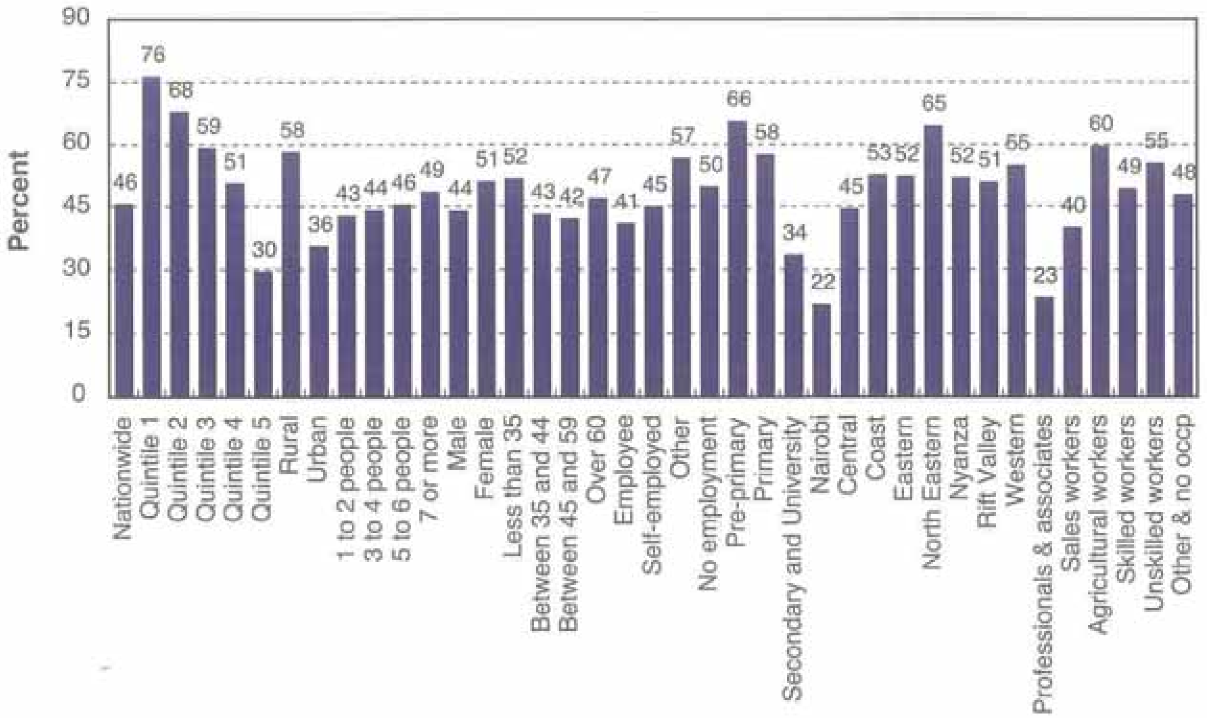
Figure 10: Monetary value of average total consumption



As shown in Figure 10, the daily average total consumption per person, which includes both food and non-food expenditures, was KSh 91.10 at the national level. Total expenditures differed according to the sub national groups of the population. Households in the highest income quintile recorded a daily average total consumption per person of KSh 351.50, while those in the lowest quintile had the least value of KSh 22.60. Small sized households registered a daily average total consumption per person of KSh 235.10 while larger households recorded KSh 59.00. Households' whose heads' had higher education spent more on food and non-food items per person with a total consumption expenditure of KSh 173.20. Average total consumption by urban households was almost three times that of rural households, KSh 173.80 and KSh 57.40 respectively. Regionally, households in Nairobi daily average total consumption per person was about KSh 482.20 while the rest of the regions spent between KSh 44.80 in North Eastern province and KSh 108.10 in Central province.

The share of food expenditure to total consumption known as Engel ratio is an indicator of consumption pattern, and usually is closely related to income. Nationally, Engel's ratio was reported as 46 per cent. The lowest income quintile as shown in Figure 11 registered the highest amount of expenditure (76 per cent) on food consumption in relation to the total household consumption. The finding is consistent with Engel's law which states that the proportion of money a household spends on food decreases with increasing income.

Figure 11: Share of food monetary value to total consumption



Households in rural areas spent most of their income on food compared to urban households with respective shares of 58 and 36 per cent. Regionally, it was observed that households in Nairobi on average spent 21.8 per cent of their income on food compared to other provinces that spent about 55 per cent of their income to acquire food.

Figure 12 depicts the trend of average food consumption compared to average total consumption for deciles of income levels. The gap between the total consumption and food expenditure for the lowest income deciles was small and increased along with increase of income.

Figure 13 summarizes the share of food monetary value to total consumption by income decile. The lower income decile spent much of their income on acquisition of food while the higher income decile spent only small fraction of their expenditure on food. This is because the lower income group does not have a lot of resources at their disposal to spend on other goods and services.

Figure 12: Average food expenditure and average total expenditure

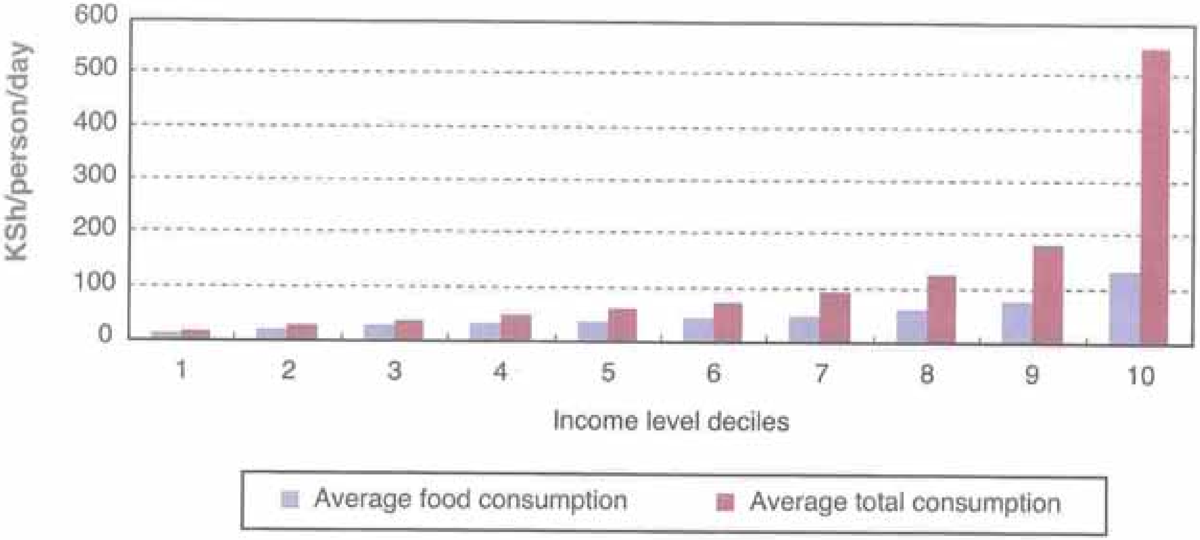
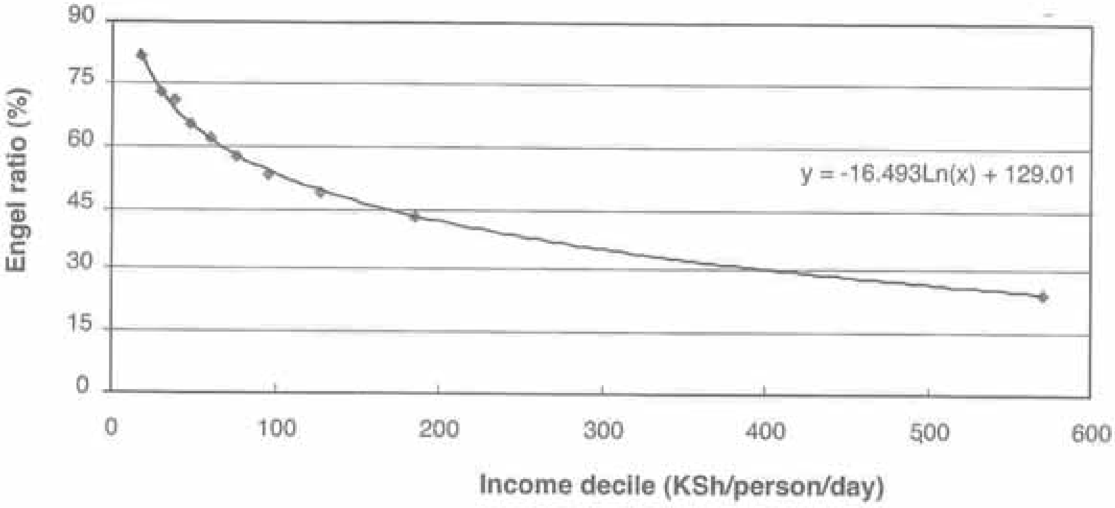


Figure 13: Share of food monetary value to total consumption



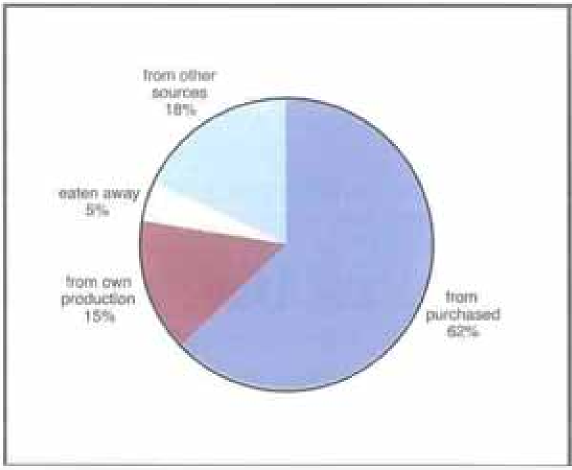
3.4.4. Share of food consumption by food source

Most of the food consumed by the population was acquired through purchases, own production or obtained free. Some people consumed outside their homes in restaurants or from street vendors. At the national level, as shown in Figure 14a, households acquired 62 per cent of their food from purchases, 15 per cent from their own production, 5 per cent eaten away from home while 18 per cent from other sources. This therefore means that majority of households did not rely on subsistence farming as their main source of food. The most likely reason could be they were engaged in income generating activities. Majority of the households that relied on eating away were mostly found in urban centres.

In terms of dietary energy as shown in Figure 14b, the source of acquisition are the same but the shares differed slightly as purchased accounted for 58 per cent of dietary energy consumed while other sources accounted for more than one fourth of dietary energy consumed and own production contributed only 13.5 per cent of the dietary energy consumed.

Figure 14a. Food consumption in monetary value

Figure 14b. Food consumption in dietary energy



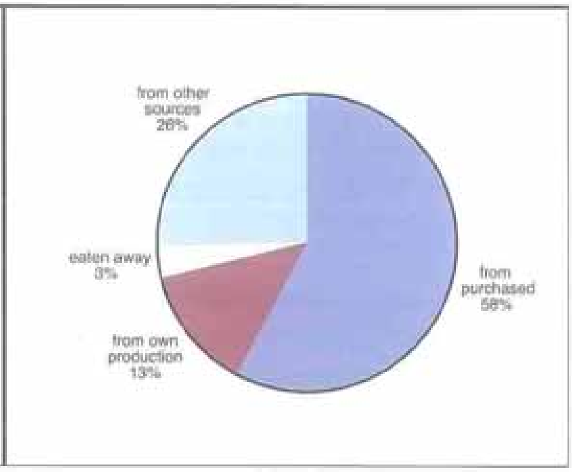
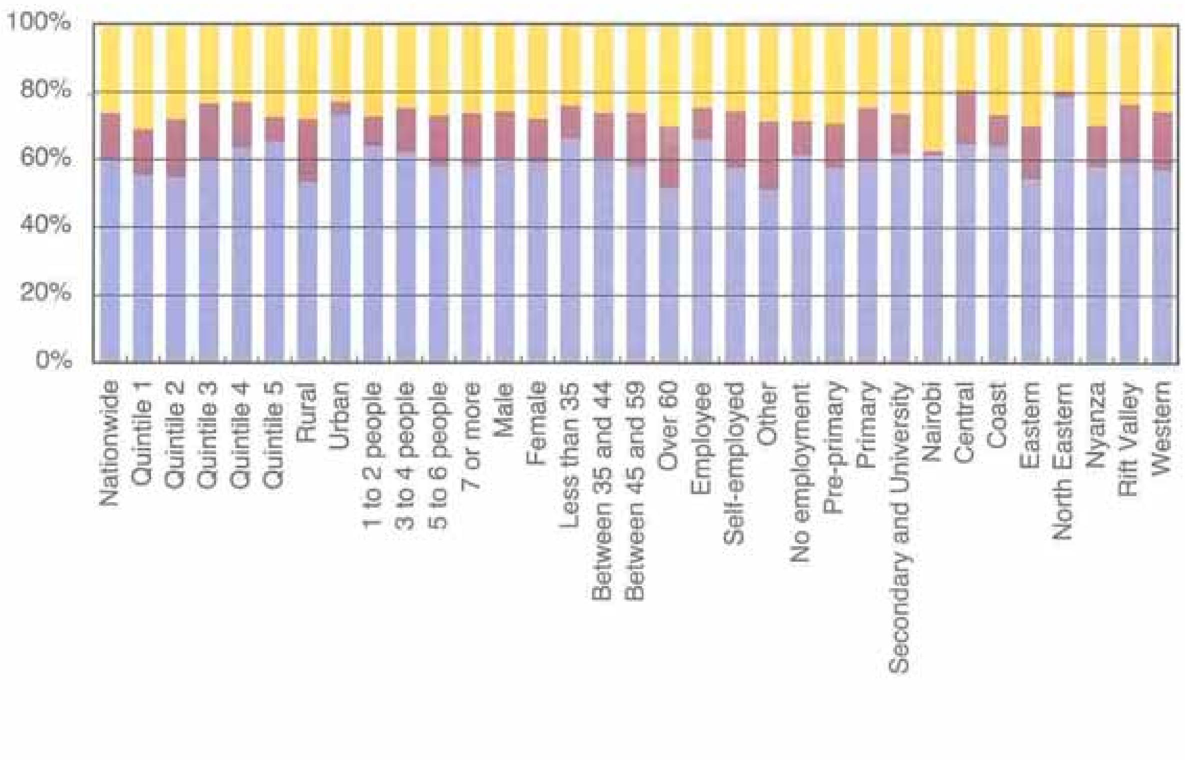


Figure 15 shows the sources of acquisition of dietary energy by sub national groupings. Households in urban and rural are'as acquired 52.4 per cent and 68.4 per cent respectively of their dietary energy from purchases. In addition, rural households derived 18.3 per cent of their dietary energy from own production while urban households derived only 3.6 per cent. Own production remained a fairly important food source for most of the population groupings. This is mostly due to the availability of land for cultivation. Food eaten away from home was almost insignificant among most population groupings.

Figure 15: Share of dietary energy consumption by food sources



3.5 Dietary Diversity

Dietary diversity is a qualitative measure of food consumption that reflects household access to a variety of foods and acts as proxy for nutrient adequacy for individual diets. A simple count of foods consumed over a reference period (mostly over 24 hour period) is made. A high score is associated with increased nutrient adequacy of the diet. A score below three reflects poor dietary diversity and is unacceptable, a score of three is regarded acceptable while a score of four and above reflects adequate nutrient intake. The measure also shows the economic ability of a household to consume a variety of foods. Studies have shown that high dietary diversity score is associated with socio-economic status and household food security. In case of Kenya, food items collected in the KIHBS were grouped into 19 different categories of food products to identify those food groups that contributed the most to the diet of the Kenyan in terms of quality and quantities of nutriments consumed. Figure 16 shows that on average, dietary energy consumption consisted of 12 per cent of proteins, 21 per cent of fats and 62 per cent of carbohydrates. Kenya is fairly within the recommended World Health Organization standard for a balanced diet as shown in Figure 17, consisting of 10-15 per cent of proteins, 15-30 per cent of fat and finally 55-75 per cent of carbohydrates. Carbohydrates are relatively cheap and easily available to households while cost of proteins and fats tend to be prohibitive hence not easily accessible.

The share of nutrients to dietary energy consumption as shown in Figure 18 depicts that carbohydrates were a major food nutrient to DEC in all categories. The diet remained balanced for all categories; in particular, the share of carbohydrates in urban and rural areas was 65 and 68 per cent respectively. The share of proteins was 12 per cent in both areas while that of fats was higher in urban areas than in rural areas with a respective contribution of 23 and 20 per cent to the DEC. Regionally, North Eastern province had the least consumption of proteins of about 10 per cent while Nyanza had the highest consumption of fats of about 25 per cent.

Figure 19 shows the per centage contribution of dietary energy consumption by main food items groups. Cereals and cereal products constituted the main dietary energy supply by contributing 47.8 per cent followed by pulses at 9.4 per cent, sugar and syrup products at 9.2 per cent.

Figure 16: Share of nutrients to dietary energy consumption

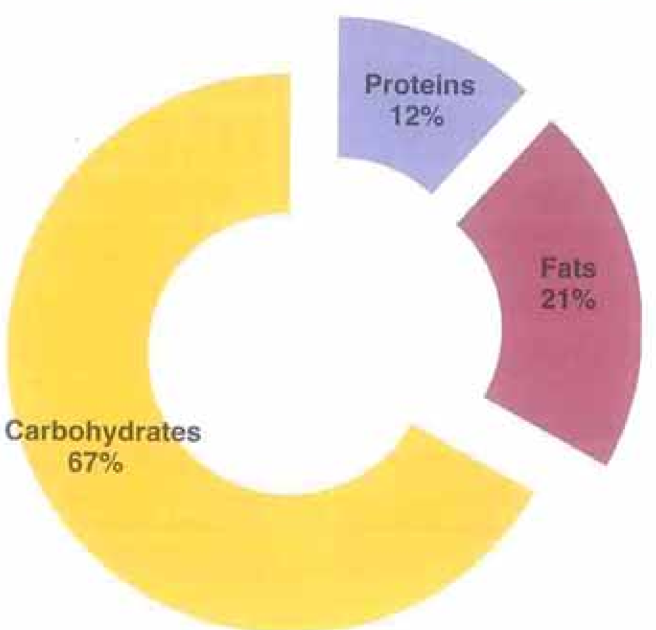


Figure 17: Comparison of share of nutrients to Dietary Energy Consumption with WHO norms

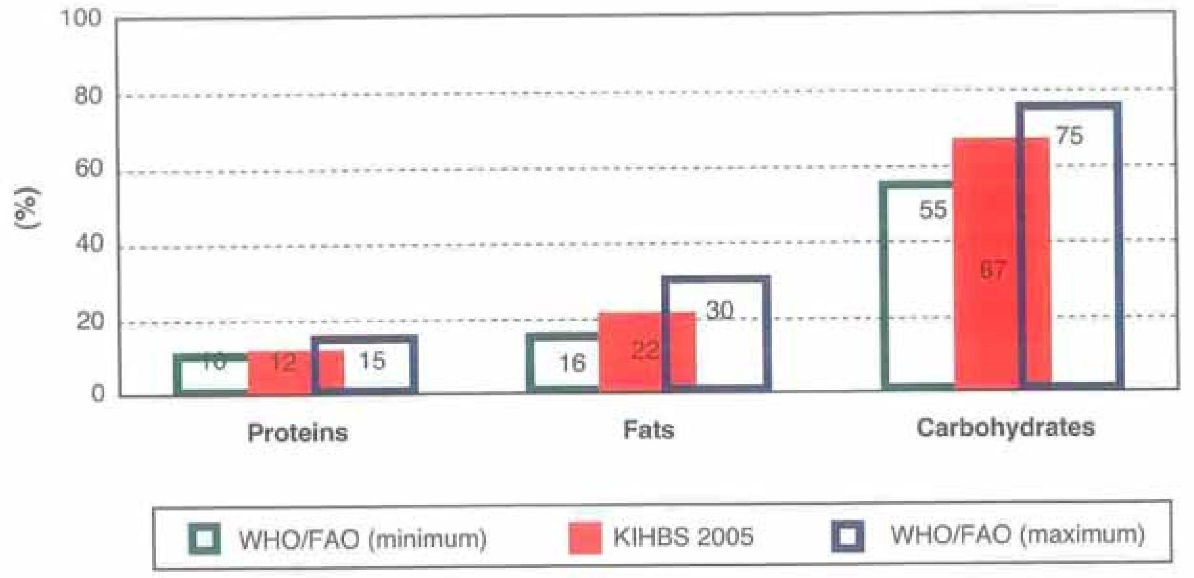


Figure 18: Share of nutrients to DEC by sub national groupings

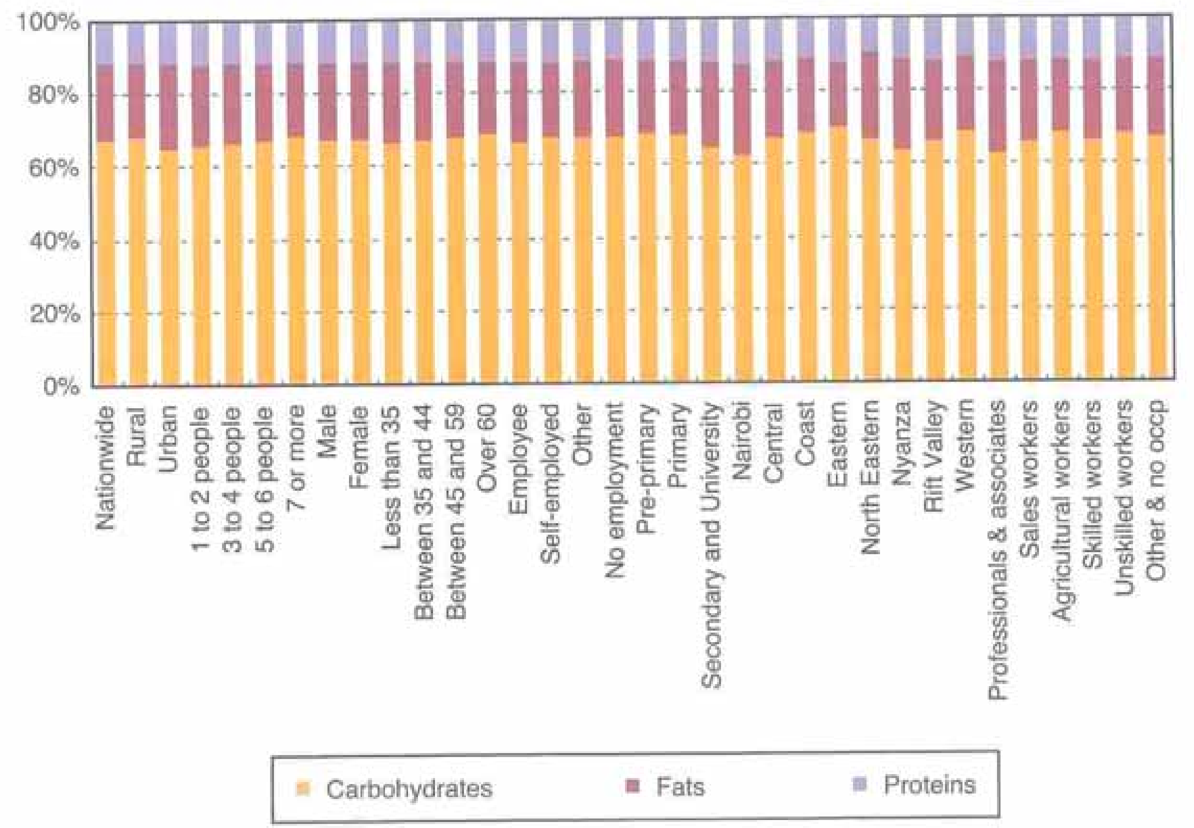


Figure 19: Share of dietary energy consumption in total energy consumption

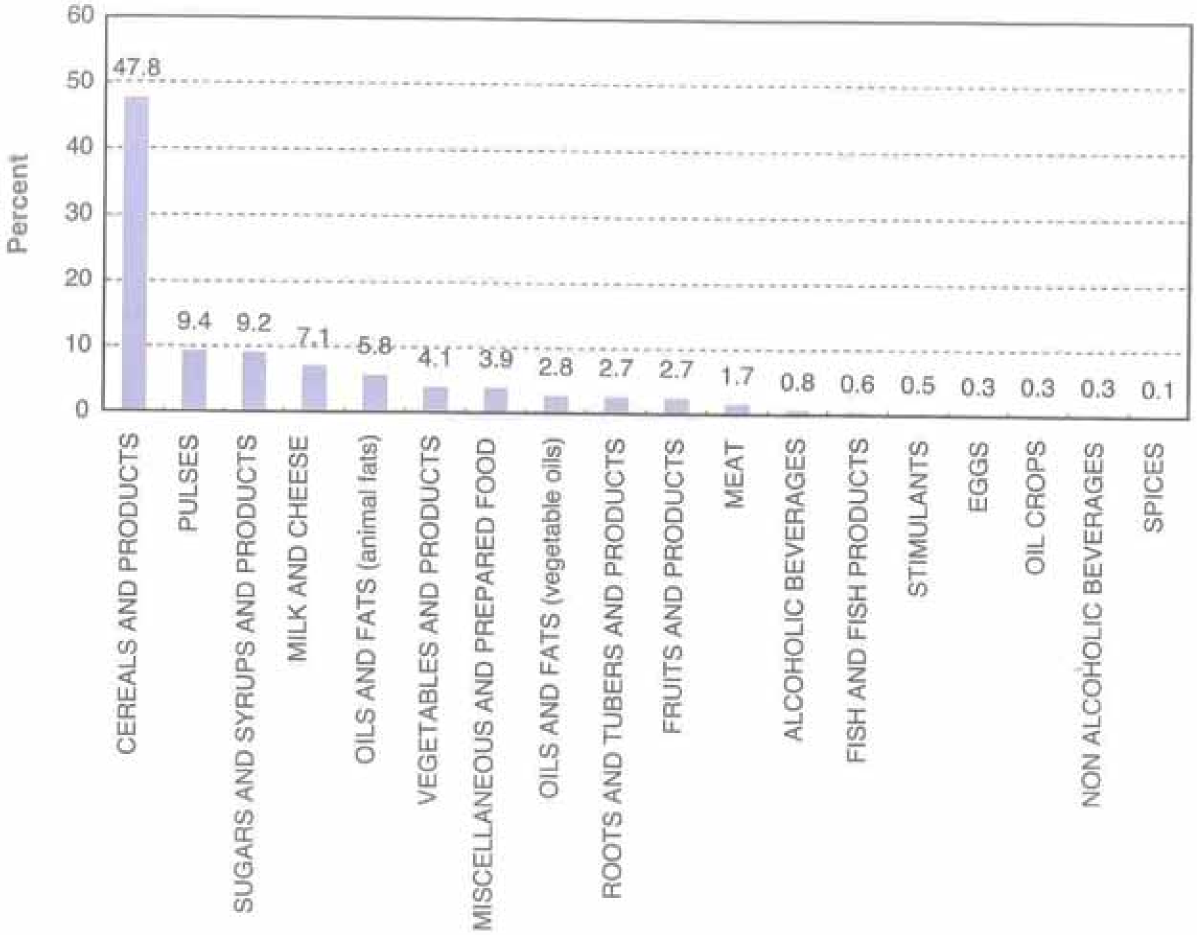


Figure 20 depicts the nutrient contribution of each food groups to total nutrient consumption. Cereals are main providers of carbohydrates followed by pulses. Proteins obtained from vegetables and meat, dairy products and fish accounted for less than 24 per cent of total proteins consumed. Animal fats were consumed more than vegetable fats. Despite Kenya having a balanced diet in terms of proteins, fats and carbohydrates consumption with respect to WHO/FAO/UNU recommendations, most of the nutrients came from vegetables. This is mainly attributed to the higher unit value of meat and fish compared to that of cereals as shown in Table 1. In order to obtain a thousand kcal from cereals one needed to spend on average KSh 11.70 while meat and fish the cost was much higher, KSh 155.50 and KSh 87.50 respectively. The unit cost of nutrients by food commodity groups showed large disparities within and between nutrients. It is noted that the large unit values were due to low nutrient values in corresponding items. The cheapest dietary energy was obtained from cereals and cereal products, oils and fats (animal fats) that went for KSh 11.00 per dietary energy unit value, oils fats (vegetable fats), pulses that cost KSh 14.00 and KSh 15.00. Spices, non-alcoholic beverages and meat were the most expensive costing KSh 287.00, KSh 161.00 and KSh 160.00 respectively.

3.6 Food and Income inequality

Inequality in access to food is estimated by FAO through the Coefficient of Variation (CV) of dietary energy consumption. This CV results into two components that capture the variation of DEC owing to income and the variation of DEC owing to energy requirement. The Coefficient of Variation (CV) of DEC due to income as shown in Figure 21 was 35 per cent resulting to an overall CV of DEC of 41 per cent including CV due to energy requirement.

Figure 20: Per centage contribution of dietary energy by main food items

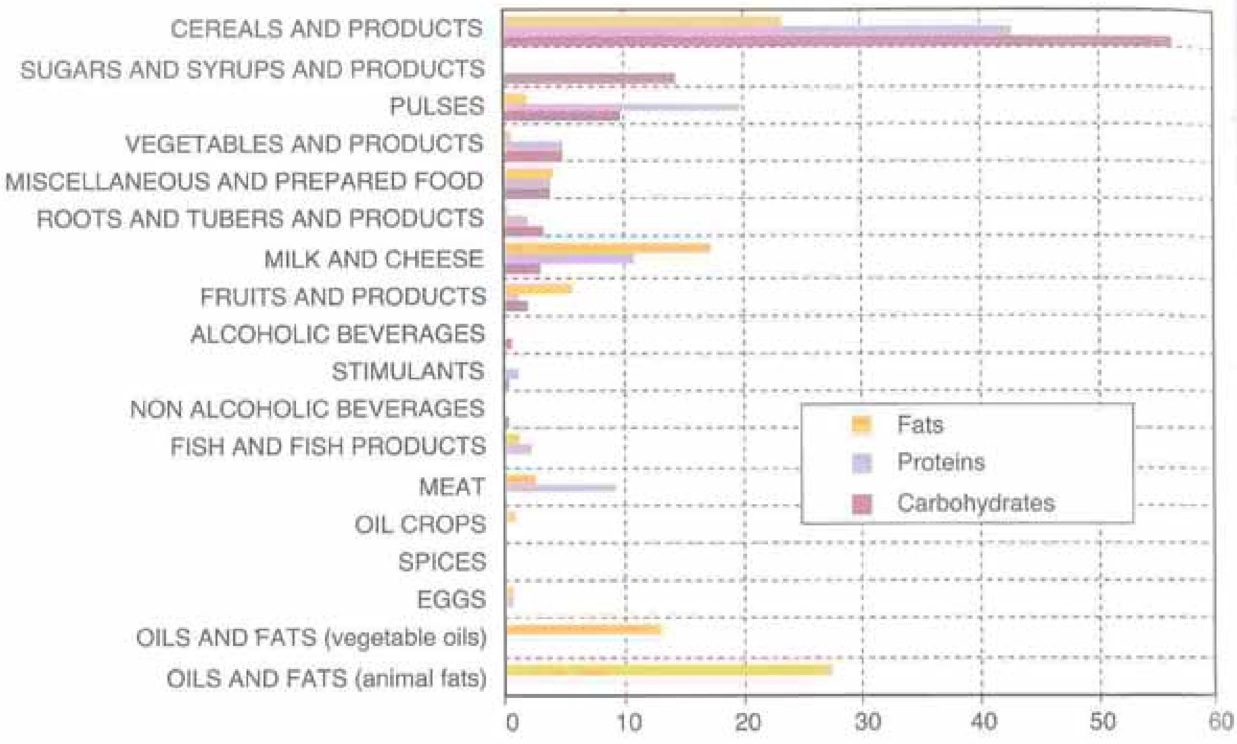


Table 1: Nutrient costs by food commodity groups

Food commodity group	Average Dietary Energy Unit Value (KSh/1000kcals)	Proteins Unit Value (KSh/100g)	Carbohydrates Unit Value (KSh/100g)	Fats Unit Value (KSh/100g)
CEREALS AND PRODUCTS	11.27	42.48	5.61	97.41
OILS AND FATS (animal fats)	11.31	0.00		10.17
OILS AND FATS (vegetable oils)	13.59			12.24
PULSES	14.09	22.54	7.83	271.10
SUGARS AND SYRUPS AND PRODUCTS	20.36		7.66	
OIL CROPS	28.80	164.47	42.77	36.82
ROOTS AND TUBERS AND PRODUCTS	29.08	139.25	14.49	
MISCELLANEOUS AND PREPARED FOOD	33.38	116.31	20.16	133.13
MILK AND CHEESE	35.49	77.24	47.36	61.32
FRUITS AND PRODUCTS	44.92	385.82	34.61	88.61
VEGETABLES AND PRODUCTS	59.44	163.03	29.32	
ALCOHOLIC BEVERAGES	78.71		61.01	0.00
FISH AND FISH PRODUCTS	87.45	79.18	182.01	212.95
STIMULANTS	97.35	142.40	59.34	
EGGS	103.81	126.82	561.13	157.97
MEAT	155.52	96.96		477.11
NON ALCOHOLIC BEVERAGES	177.69		73.91	
SPICES	282.78		154.08	

Figure 21: National wide coefficients of variation of food consumption and income

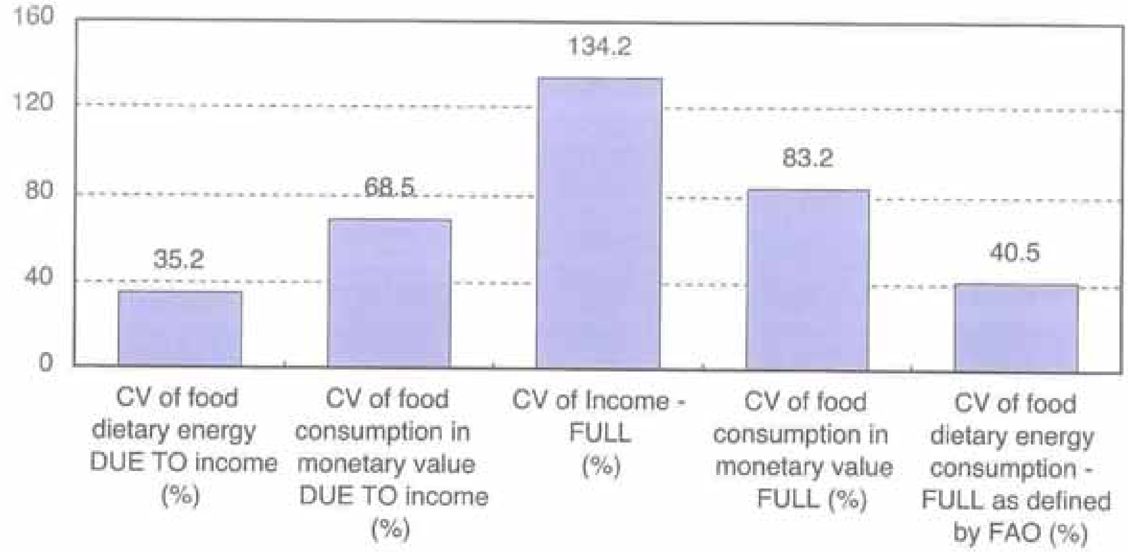
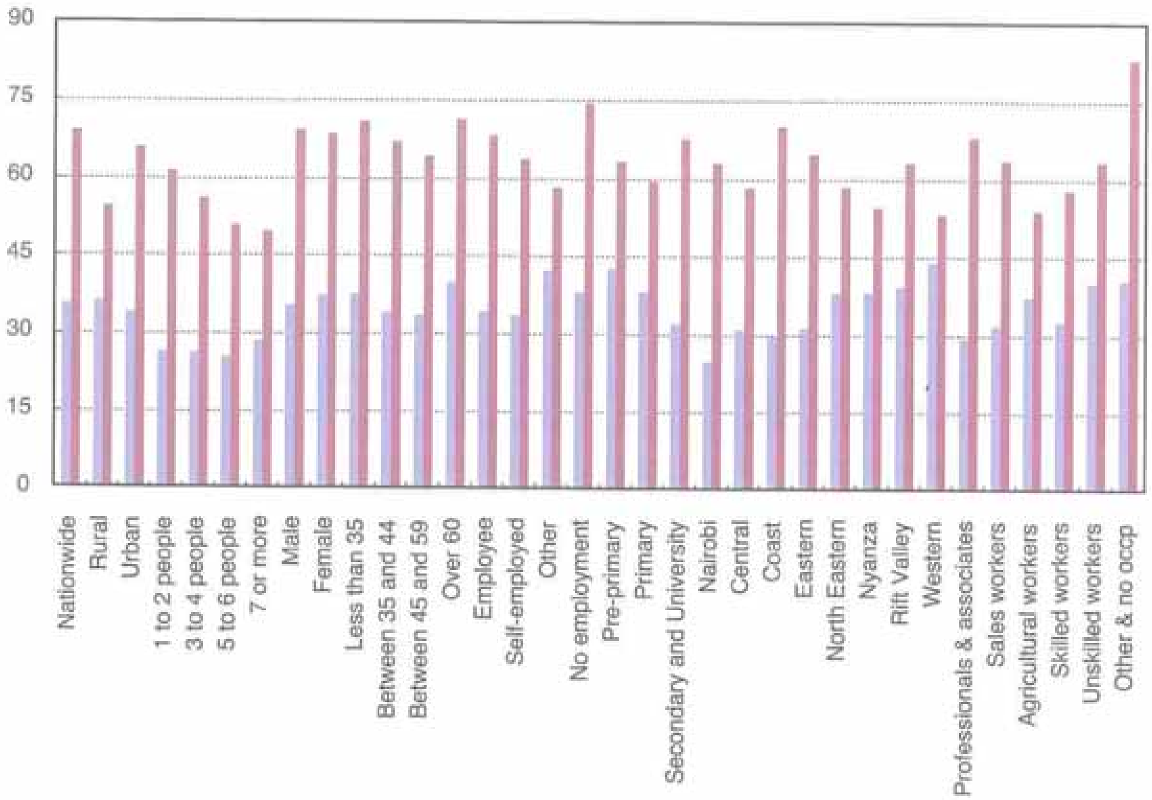


Figure 22: Inequality of food consumption due to income by Sub-national levels



The CV of food consumption in monetary value due to income was 69 per cent compared to dietary energy which registered 35 per cent as the former includes the variations of food prices. CV of income commonly called CV income FULL was 134 per cent compared to the CV of DEC.

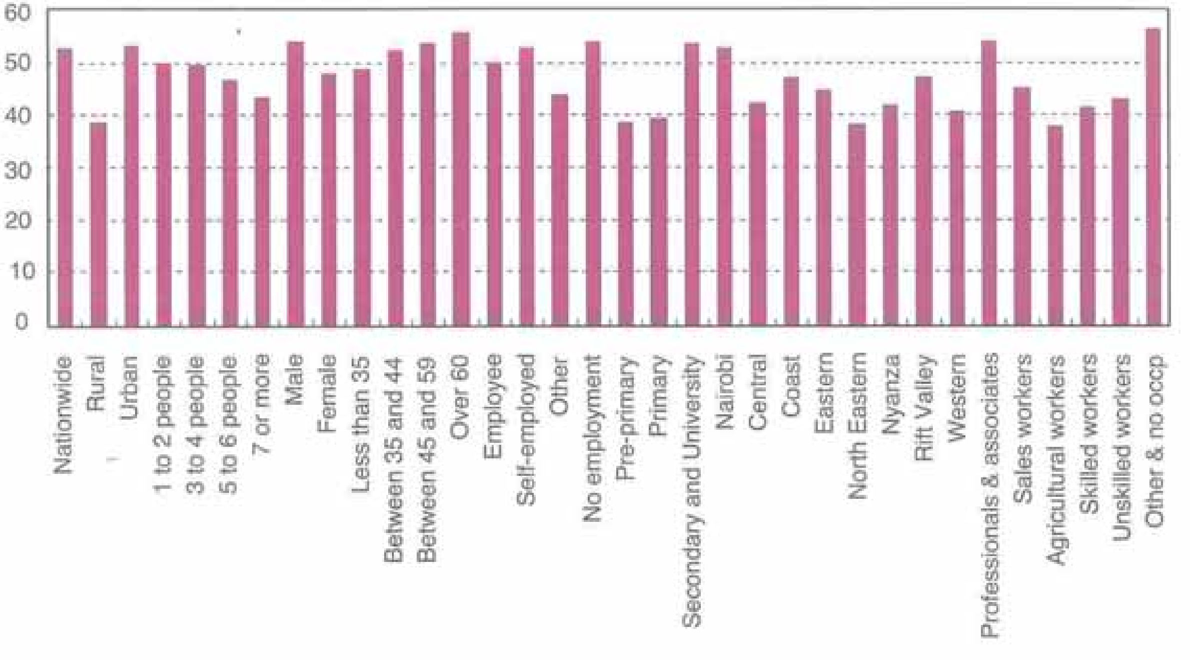
3.6.1. Inequality in access to food consumption due to income at sub national levels

Inequality in access to food as measured by the CV of DEC due to income as shown in Figure 22 was 35 per cent nationwide but inequalities differed according to the sub groups of population. Higher inequalities of DEC of magnitude more than the national level were observed in rural areas, female headed households, household heads with less than 35 and over 60 years, heads without employment and higher education. Male headed households, households whose head was less than 35 and over 60 years and without employment had more CV of food consumption in monetary value due to income than national level of 68.5 per cent.

3.6.2. Income inequality

The Gini coefficient of total consumption due to income which measures the disparity of income, as shown in Figure 23 was 52 per cent at national level. Higher inequalities were observed in urban areas, male headed households, household heads between 35 and 60 years, heads without employment and those having higher education.

Figure 23: GINI of Income - FULL (%) by Sub-national levels



3.6.3. Dispersion (inequality) ratios

The dispersion ratio of a variable measures the disparity between the lowest and highest quintile (80/20). It is estimated for the food consumption in both monetary and dietary energy values, consumption and income. Figure 24 shows that at national level, the last quintile was spending on food six times more than the first quintile and consuming three times more in terms of dietary energy consumption. Income inequality was also wide as income of the last quintile was sixteen times higher than that of the first quintile. The variability in dietary energy consumption is less than food expenditure as this latter includes additional variability of food prices. Total household consumption includes variability of all commodities by types and prices. Variations in income include many components of socio economic factors. Dispersion ratios estimated for various sub national levels for each of the four components are given in the annex tables.

Figure 24: Dispersion Ratios (Inequality) at National Level

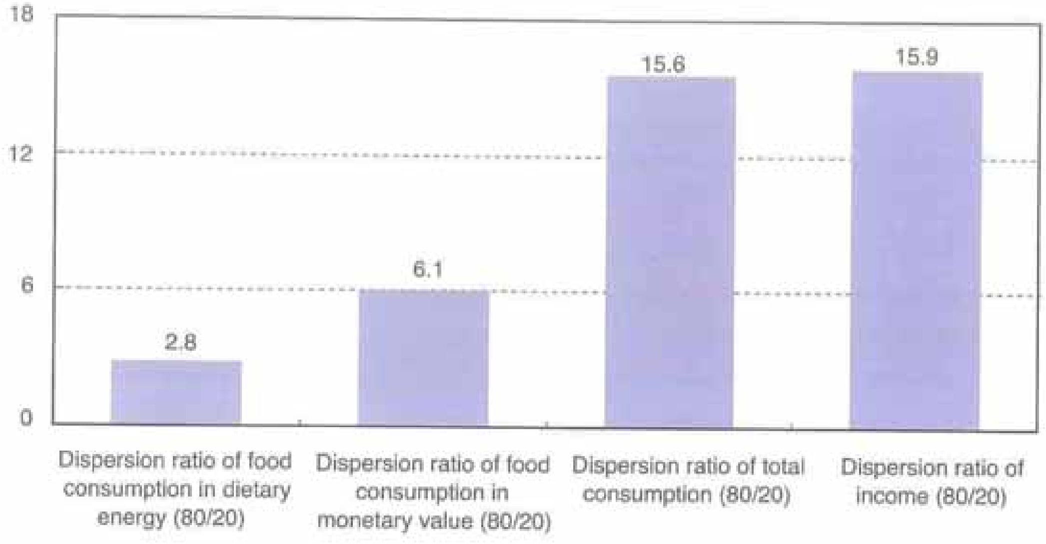
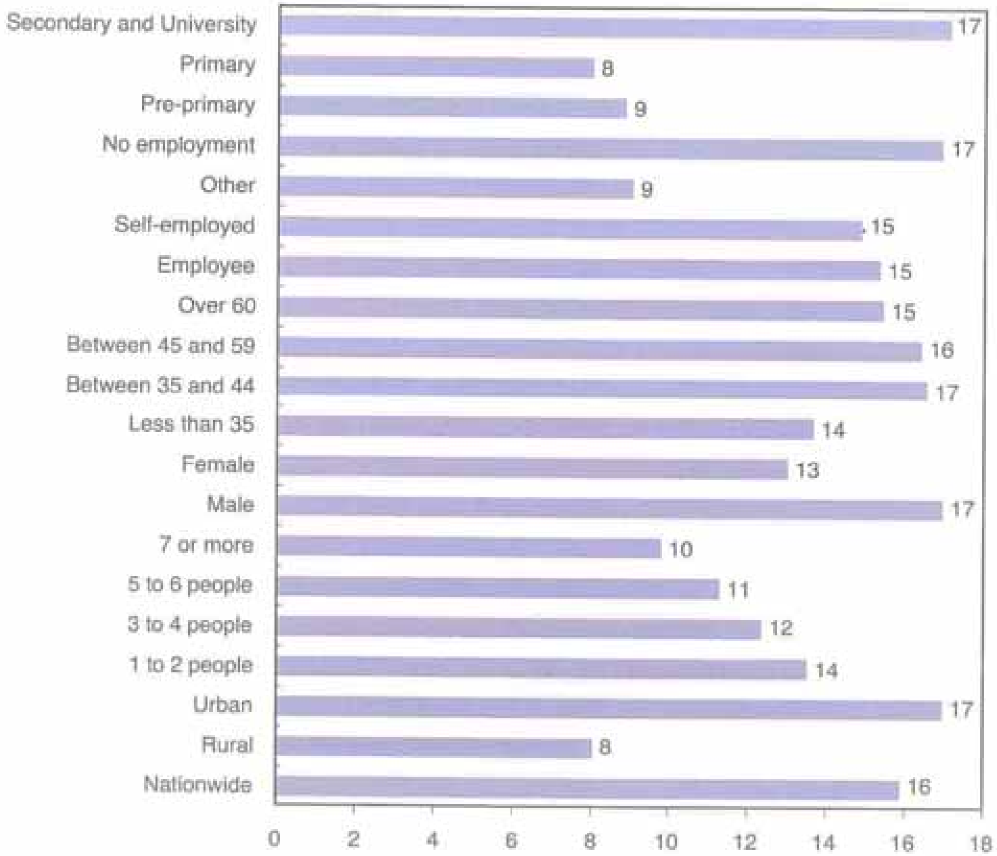


Figure 25 shows that the highest inequality in income was observed within male headed households who had higher education and mainly located in urban areas. The households' had income of the last quintile being 17 times higher than income of the first quintile. While rural households whose head had primary level of education had the least income inequalities. The last quintile had income 8 times that of the first quintile.

Figure 25: Dispersion ratio of income (80/20) (2005/06 KIHBS)



3.7 Towards the WFS and MDG

More than ten years ago, world leaders met in Rome for the World Food Summit (WFS) to discuss ways to end hunger. They pledged their commitments to an ongoing effort to eradicate hunger in all countries and set themselves the target of halving the number of undernourished people by 2015. Countries also committed to the Millennium Development Goal (MDG 1) of halving hunger and extreme poverty by the year 2015.

Figure 26 shows that the estimate of the MDG indicator 5 from the 2005/06 KIHBS was 51 per cent. This was above the projected FAO 2001-03 value of 31 per cent as published in the State of Food Insecurity 2006. According to FAO global estimates, the per centage of the undernourished reduced from 39 per cent in 1990-92 to 31 per cent in 2001-03.

Figure 26: Progress towards MDG

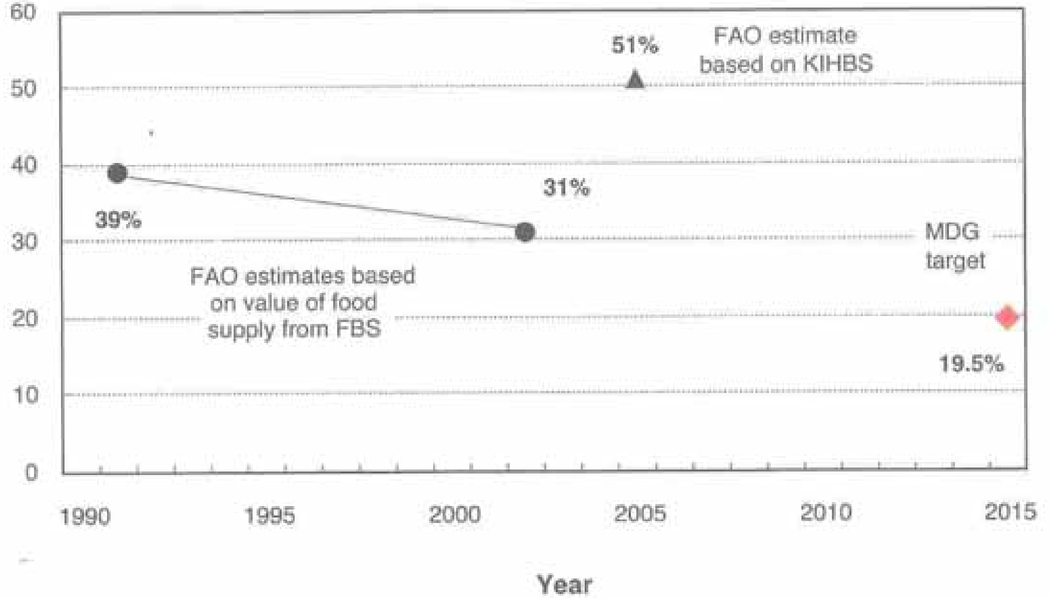


Figure 27: Progress towards WFS

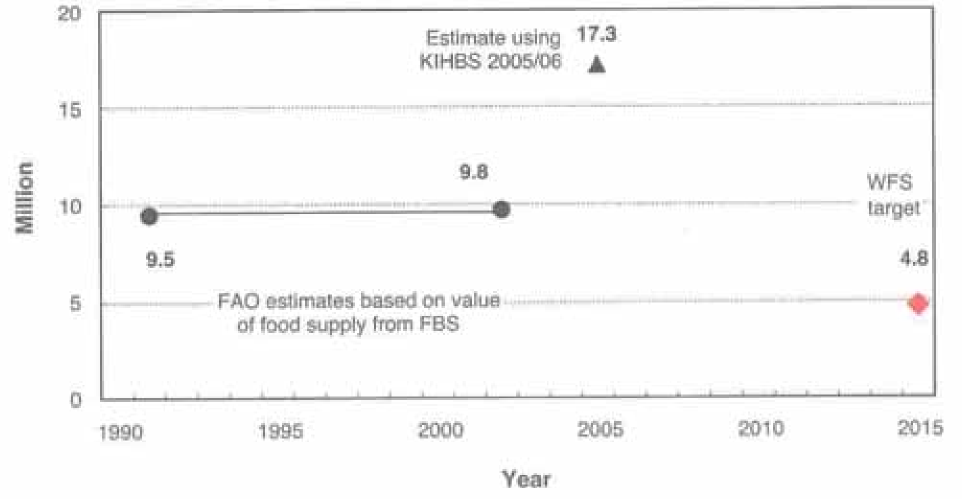


Figure 27 depicts the progress the country has made towards achieving the WFS target of reducing the food deprived to about 5 million by 2015. The country's estimate using the 2005/06 KIHBS showed that 17 million people were undernourished. As for the prevalence of food deprivation, it necessary to use the available 2005/06 KIHBS food consumption data to analyse the trends over time at both national and sub national levels.

CHAPTER 4

CONCLUSION

This report analyses the food security situation in Kenya using data on food consumption from the 2005/06 Kenya Integrated Household Budget Survey, which is an important source of data for the derivation of food security indicators at national and sub national levels. The annual estimation of the MDG indicator 5 can be performed for the assessment and monitoring of the food situation at the national and sub national levels.

KNBS has a good statistical infrastructure for collecting household food consumption data and can easily use the Food Security Statistics Module (FSSM) to derive Food Security Statistics (FSS) at the national and sub national levels and over time. The FSS are vital for the design, implementation, monitoring and evaluation of anti hunger policies for more effective targeted interventions. The reliability and accuracy of the estimates are dependent on the accuracy of the KIHBS collecting food consumption data and measuring income and expenditure distribution across the population. Below are some limitations that have been encountered while performing the food security analysis of the 2005/06 KIHBS. It is important to implement the recommendations to increase the accuracy of future surveys that would provide data for deriving FSS estimates over regions and time.

4.1 Limitations

The 2005/06 KIHBS food consumption data made available had some inconsistencies, such as:

- Wrongly entered units of quantity measurement;
- Very big dispersion of purchased price for food items;
- No monetary values for some purchased items;
- Incomplete information on some demographic characteristics of households;
- Own consumption of food items included also food production from their harvest;
- Reported bulk food purchases that were meant for stock, and
- Country does not have an updated official food composition table.

4.2 Recommendations

The 2005/06 KIHBS was a very comprehensive survey. It captured a wide range of household data which was essential for updating the diaries for the proper collection of household food data. The data included consumption and acquisition easily identified, purchases, own consumption, food obtained from relatives or as gifts and own old stocks. This will enable us to have more accurate food data that is actually available and consumed at the household level.

A more elaborate editing procedure should be established in the field at the collection stage to avoid missing or abnormal prices and quantities. Further edits should be extended at the data capture stage.

There is need to prepare comprehensive nutrient conversion factors and update food composition table for Kenya which will be useful for the FSSM and for extensive food micronutrients analysis such as vitamins and iron, among others.

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GLOSSARY OF TERMS

ANTHROPOMETRY
Use of human body measurements to obtain information about nutritional status.

AVERAGE ENERGY REQUIREMENT
It refers to the amount of energy considered adequate to meet the energy needs for normative average acceptable weight for attained height while performing moderate physical activity in good health.

BALANCED DIET
A diet is balanced when is judged to be consistent with the maintenance of health in a population. The balance can be examined in terms of the presence of the various energy-supplying macronutrients and other nutrients. A macronutrient-based balance food consumption pattern should contribute to total energy from proteins, fats and carbohydrates within recommended ranges as follows: proteins from 10 to 15 per cent, fats from 15 to 30 per cent and carbohydrates from 55 to 75 per cent as from a technical report of a 2002 Joint WHO/FAO Expert Consultation (WHO 2003).

CRITICAL FOOD POVERTY
The prevalence of critical food poverty (pCFP) refers to the proportion of persons living on less than the cost of the macro-nutrient balanced MDER (for Minimum Dietary Energy Requirement see below and for balanced diet see above) with food prices from households in the lowest income quintile. It can be estimated at national and sub-national levels.

DEGREE OF FOOD DEPRIVATION
A measure of the overall food insecurity situation in a country, based on a classification system that combines prevalence of undernourishment, i.e. the proportion of the total population suffering from from a dietary energy deficit, and depth of undernourishment, i.e. the magnitude of the undernourished population's dietary energy deficit.

DEPTH OF FOOD DEPRIVATION
It refers to the difference between the average dietary energy intake of an undernourished population and its average minimum energy requirement (MDER).

DIETARY ENERGY UNIT COST
The dietary energy unit cost is the monetary value in local currency of 1000 kilo-calories of food consumed.

DIETARY ENERGY CONSUMPTION
This is food consumption expressed in energy terms. At national level, it can be calculated from the FBS; the FBS estimate refers to both private (households) and public (hospitals, prisons, military compounds, hotels, residences, etc) food consumption. At sub-national levels is estimated using food consumption data in quantities collected in national household surveys (NHS); these estimates refer to private food consumption.

DIETARY ENERGY DEFICIT
Same as Depth of Food deprivation.

DIETARY ENERGY INTAKE
The energy content of food consumed.

DIETARY ENERGY REQUIREMENT
It refers to the amount of energy required by individuals to maintain body functions, health and normal physical activity.

DIETARY ENERGY SUPPLY

Food available for human consumption, expressed in kilocalories per person per day (kcal/person/day). At the country level, it is calculated as the food remaining for human use after deduction of all non-food consumption (exports, animal feed, industrial use, seed and wastage)

FOOD BALANCE SHEETS

Food Balance Sheets (FBS) are compiled every year by FAO, mainly with country-level data on the production and trade of food commodities. Using these data and the available information on seed rates, waste coefficients, stock changes and types of utilization (feed, food, processing and other utilization), a supply/utilization account is prepared for each commodity in weight terms. The food component of the commodity account, which is usually derived as a balancing item, refers to the total amount of the commodity available for human consumption during the year.

FOOD CONSUMPTION DISTRIBUTION

Food consumption distribution refers to the variation of consumption within a population. It reflects both the disparities due to socioeconomic factors and differences due to biological factors, such as sex, age, body weight and physical activity levels.

FOOD DEPRIVATION

Food deprivation refers to the condition of people whose food consumption is continuously below their requirements. FAO's measure of food deprivation refers to the proportion of the population whose dietary energy consumption is below the minimum energy requirement (see below).

FOOD INSECURITY

This is a situation that exists where people lack secure access to sufficient amounts of safe and nutritious food for normal growth and development and an active and healthy life. It may be caused by the unavailability of food, insufficient purchasing power, inappropriate distribution, or inadequate use of food at the household level. Food insecurity, poor conditions of health and sanitation, and inappropriate care and feeding practices are the major causes of poor nutritional status. Food insecurity may be chronic, seasonal or transitory.

FOOD SECURITY

This is a situation that exists when all people, at all times, have physical, social and economic access to sufficient, safe and nutritious food that meets their dietary needs and food preferences for an active and healthy life.

GINI COEFFICIENT

Gini index measures the extent to which the distribution of income (or, in some cases, consumption expenditure, food dietary energy consumption) among individuals or households within an economy deviates from a perfectly equal distribution. A Lorenz curve plots the cumulative per centages of total income received against the cumulative number of recipients, starting with the poorest individual or household. The Gini index measures the area between the Lorenz curve and a hypothetical line of absolute equality, expressed as a per centage of the maximum area under the line. Thus a Gini index of 0 represents perfect equality, while an index of 100 implies perfect inequality.

GINI COEFFICIENT DUE TO INCOME

The Gini coefficient is a measure of inequality in food consumption when income is used as the grouping variable and ranges from 0 (when income has no effect on food consumption) to 1 (when food consumption depends only on income). It can refer to inequality in food consumption due to income in monetary or in energy terms.

HOUSEHOLD CONSUMPTION EXPENDITURE

Household consumption expenditure refers to all money expenditure by the household and individual members on goods intended for consumption and expenditure on services, plus the value of goods and services received as income in kind and consumed by the household or individual members of the household. Thus the value of items produced by the household and utilised in its own consumption, the net rental value of owner-occupied housing and the gross rental value of free housing occupied by the household represent part of household consumption expenditure.

HOUSEHOLD FOOD CONSUMPTION EXPENDITURE

It refers to food consumed by household members during a specified period, at home and outside the home, for example, at restaurants, bars, the work place, school, and so on. It includes food from all sources, purchased or from garden or farm. Further deductions should be made to allow food given away to other households or non-household members and visits as well as for wastage and losses occurring from acquisition to cooking and plate and kitchen wastage.

HOUSEHOLD EXPENDITURE

Consumption plus non-consumption expenditure made by the household, including food.

HOUSEHOLD NON CONSUMPTION EXPENDITURE

It refers to income taxes, other direct taxes, pension and social security contributions, remittances, gifts and similar transfers made by the household in monetary terms or in kind, including food such as given away raw or ready to eat.

HOUSEHOLD INCOME

Income is the sum of all receipts, in money or in kind, which as a rule are received regularly and are of recurring nature, including food.

INCOME ELASTICITY OF FOOD DEMAND

The income elasticity of food demand measures the responsiveness of the quantity, monetary or nutrient value demanded of a good to the change in the income of the people demanding the good. It is calculated as the ratio of the per cent change in quantity demanded to the per cent change in income.

INCOME INEQUALITY

Income inequality refers to disparities in the distribution of income.

INEQUALITY IN FOOD CONSUMPTION DUE TO INCOME

The inequality refers to the variation of the food consumption level within a population due to disparities in the income distribution.

KILOCALORIE (Kcal)

Kilocalorie is a unit of measurement of dietary energy. In the International System of Units (ISU), the universal unit of dietary energy is the joule (J) but Kcal is still commonly used. One kilocalorie = 4.184 kilo-joules (KJ).

MACRONUTRIENTS

Used in this document to refer to the proteins, carbohydrates and fats that are required by the body in large amounts and that are available to be used for energy. They are measured in grams.

MICRONUTRIENTS

Refer to the vitamins, minerals and certain other substances that are required by the body in small amounts. They are measured in milligrams or micrograms.

MINIMUM DIETARY ENERGY REQUIREMENT
In a specified age/sex category, the amount of dietary energy per person that is considered adequate to meet the energy needs for maintaining a healthy life and carrying out a light physical activity. For an entire population, the minimum energy requirement is the weighted average of the minimum energy requirements of the different age/sex groups in the population. This is expressed as kilocalories per person per day.

NUTRITIONAL STATUS
The physiological status of an individual that results from the relationship between nutrient intake and requirement and from the body's ability to digest, absorb and use these nutrients. Lack of food as well as poor health and sanitation and inappropriate care and feeding practices are the major causes of poor nutritional status.

SHARE OF FOOD EXPENDITURE
The proportion of household consumption expenditure allocated to food; it is also known as Engel ratio.

UNDER NOURISHMENT
Undernourishment refers to the condition of people whose dietary energy consumption is continuously below a minimum dietary energy requirement for maintaining a healthy life and carrying out a light physical activity. The number of undernourished people refers to those in this condition.

UNDERNUTRITION
The result of undernourishment, poor absorption and/or poor biological use of nutrients consumed.

Annex Table 1: Selected Statistics on Food Consumption in Kenya

Categories and Groupings	Number of sampled households	Average number of people in household	Average food consumption in dietary energy value (kcal/person/day)	Average food consumption in monetary value (KSh /person/day)	Average dietary energy unit value (KSh/1000kcals)	Average total consumption (KSh/person/day)
Nationwide	12,980	5.0	1,800	41.73	23.24	91.09
Income level						
Quintile 1	2,596	6.8	1,080	17.21	15.92	22.55
Quintile 2	2,596	5.9	1,560	29.28	18.71	43.13
Quintile 3	2,596	5.0	1,910	40.31	21.16	67.95
Quintile 4	2,596	4.2	2,280	55.95	24.57	110.35
Quintile 5	2,596	3.0	3,020	104.33	34.51	351.49
Area						
Rural	8,382	5.5	1,690	33.42	19.81	57.41
Urban	4,598	4.1	2,060	62.13	30.11	173.76
HH Size						
1 to 2 people	2,576	1.5	3,410	101.30	29.70	235.05
3 to 4 people	3,452	3.6	2,260	56.39	24.94	127.00
5 to 6 people	3,473	5.5	1,790	40.88	22.89	89.71
7 or more	3,479	8.6	1,410	28.73	20.42	59.03
Gender of head of HH						
Male	9,108	5.2	1,790	42.43	23.77	95.97
Female	3,872	4.5	1,820	39.84	21.83	77.84
Age of head of HH						
Less than 35	4,084	3.8	2,000	50.90	25.48	97.98
Between 35 and 44	3,239	5.5	1,710	40.96	23.89	94.36
Between 45 and 59	3,363	6.1	1,690	37.46	22.15	88.61
Over 60	2,294	4.8	1,840	38.02	20.69	80.76
Economic activity of head of HH						
Employee	3,739	4.4	1,910	51.42	26.90	124.76
Self-employed	4,785	5.1	1,840	40.93	22.27	90.54
Other	1,732	5.6	1,650	35.13	21.26	61.73
No employment	2,724	5.1	1,690	36.33	21.56	72.81
Education of head of HH						
Pre-primary	3,324	5.4	1,660	30.65	18.46	46.72
Primary	5,634	5.0	1,760	38.00	21.62	65.82
Secondary and University	4,022	4.6	1,990	58.28	29.33	173.21
Region						
Nairobi	632	3.8	2,530	105.22	41.66	482.17
Central	1,460	4.1	2,110	48.37	22.91	108.10
Coast	1,234	5.3	1,960	42.61	21.79	80.96
Eastern	2,354	5.2	2,240	37.22	16.58	71.11
North Eastern	497	6.2	1,460	28.91	19.81	44.77
Nyanza	2,086	4.7	1,490	43.92	29.41	84.52
Rift Valley	3,236	5.2	1,600	37.07	23.17	72.65
Western	1,481	5.4	1,440	36.01	25.04	65.30
Occupation						
Professionals & associates	753	4.8	2,150	76.77	35.74	328.36
Sales workers	1,656	4.5	2,020	54.45	26.95	135.70
Agricultural workers	4,561	5.5	1,690	33.59	19.89	56.21
Skilled workers	1,243	4.5	1,880	49.15	26.17	99.44
Unskilled workers	2,737	4.7	1,850	40.69	22.05	73.32
Other & no occp	2,030	5.1	1,660	37.23	22.37	77.53

KIHBS 2005/06

Annex Table 2: Food Deprivation and Parameters by Population Groupings in Kenya

Categories and Groupings	Number of sampled households	CV (%) of food dietary energy consumption (kcal/person/day) -FULL as defined by FAO	Minimum dietary energy requirement (kcal/person/day) as defined by FAO	Average of food dietary energy consumption (kcal/person/day)	Proportion of food deprivation in total population (%) as defined by FAO	Dietary energy consumption in food deprived population (kcal/person/day)
Nationwide	12,980	40.0	1,683	1,800	51	1,261
Income level						
Quintile 1	2,596	20.0	1,683	1,080	99	1,074
Quintile 2	2,596	20.0	1,683	1,560	68	1,395
Quintile 3	2,596	20.0	1,683	1,910	30	1,492
Quintile 4	2,596	20.0	1,683	2,280	8	1,543
Quintile 5	2,596	20.0	1,683	3,020	0	1,582
Area						
Rural	8,382	41.0	1,670	1,690	57	1,224
Urban	4,598	39.0	1,733	2,060	39	1,356
HH Size						
1 to 2 people	2,576	33.0	1,982	3,410	6	1,737
3 to 4 people	3,452	33.0	1,694	2,260	23	1,425
5 to 6 people	3,473	32.0	1,655	1,790	46	1,324
7 or more	3,479	34.0	1,659	1,410	75	1,185
Gender of head of HH						
Male	9,108	40.0	1,692	1,790	52	1,265
Female	3,872	42.0	1,657	1,820	48	1,241
Age of head of HH						
Less than 35	4,084	42.0	1,570	2,000	35	1,223
Between 35 and 44	3,239	39.0	1,641	1,710	53	1,234
Between 45 and 59	3,363	39.0	1,776	1,690	62	1,294
Over 60	2,294	44.0	1,724	1,840	52	1,258
Economic activity of head of HH						
Employee	3,739	39.0	1,693	1,910	45	1,302
Self-employed	4,785	39.0	1,679	1,840	48	1,284
Other	1,732	46.0	1,678	1,650	60	1,172
No employment	2,724	43.0	1,682	1,690	58	1,214
Education of head of HH						
Pre-primary	3,324	47.0	1,675	1,660	60	1,171
Primary	5,634	43.0	1,667	1,760	53	1,224
Secondary and University	4,022	37.0	1,716	1,990	41	1,348
Region						
Nairobi	632	31.0	1,745	2,530	15	1,502
Central	1,460	36.0	1,731	2,110	35	1,388
Coast	1,234	35.0	1,680	1,960	39	1,340
Eastern	2,354	37.0	1,695	2,240	27	1,384
North Eastern	497	43.0	1,595	1,460	66	1,112
Nyanza	2,086	43.0	1,666	1,490	68	1,150
Rift Valley	3,236	44.0	1,675	1,600	63	1,179
Western	1,481	48.0	1,639	1,440	70	1,082
Occupation						
Professionals & associates	753	35.0	1,751	2,150	33	1,420
Sales workers	1,656	37.0	1,708	2,020	39	1,350
Agricultural workers	4,561	42.0	1,676	1,690	57	1,218
Skilled workers	1,243	38.0	1,694	1,880	46	1,309
Unskilled workers	2,737	44.0	1,662	1,850	49	1,226
Other & no occp	2,030	45.0	1,683	1,660	59	1,191

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Annex Table 3: Share of Food Consumption to Total Consumption in Monetary Value and by Food Sources in Kenya

Categories and groupings	Number of sampled households	Share of food consumption in monetary value to total consumption (%)	Share of food consumption in monetary value purchased to total food value (%)	Share of food consumption in monetary value from own production to total food value (%)	Share of food consumption in monetary value eaten away from home to total food value (%)	Share of food consumption in monetary value from other sources to total food value (%)
Nationwide	12,980	45.8	62.7	14.7	4.6	18.1
Income level						
Quintile 1	2,596	76.3	57.2	19.0	0.7	23.0
Quintile 2	2,596	67.9	57.2	23.0	1.5	18.3
Quintile 3	2,596	59.3	62.7	19.7	2.1	15.5
Quintile 4	2,596	50.7	66.9	14.4	3.5	15.2
Quintile 5	2,596	29.7	64.7	5.4	9.9	19.9
Area						
Rural	8,382	58.2	55.9	23.2	1.9	19.0
Urban	4,598	35.8	71.7	3.4	8.1	16.8
HH Size						
1 to 2 people	2,576	43.1	64.3	6.8	11.3	17.7
3 to 4 people	3,452	44.4	65.2	11.9	5.2	17.7
5 to 6 people	3,473	45.6	61.7	16.1	3.2	18.9
7 or more	3,479	48.7	60.9	19.0	2.2	17.9
Gender of head of HH						
Male	9,108	44.2	62.7	14.7	5.2	17.4
Female	3,872	51.2	62.7	14.6	2.7	19.9
Age of head of HH						
Less than 35	4,084	51.9	68.1	9.8	6.5	15.6
Between 35 and 44	3,239	43.4	64.0	13.6	4.5	17.9
Between 45 and 59	3,363	42.3	60.7	17.2	4.0	18.2
Over 60	2,294	47.1	54.0	21.0	2.1	22.8
Economic activity of head of HH						
Employee	3,739	41.2	68.2	8.1	6.6	17.1
Self-employed	4,785	45.2	60.4	17.7	4.3	17.7
Other	1,732	56.9	54.4	25.1	2.1	18.4
No employment	2,724	49.9	63.9	12.5	3.2	20.4
Education of head of HH						
Pre-primary	3,324	65.6	59.5	18.0	1.6	20.9
Primary	5,634	57.7	62.7	17.8	3.4	16.1
Secondary and University	4,022	33.6	64.4	9.7	7.2	18.7
Region						
Nairobi	632	21.8	58.0	0.9	14.3	26.7
Central	1,460	44.7	63.3	16.8	3.2	16.7
Coast	1,234	52.6	64.8	11.3	4.3	19.6
Eastern	2,354	52.3	59.3	18.2	3.4	19.1
North Eastern	497	64.6	80.7	4.9	0.5	13.9
Nyanza	2,086	52.0	64.6	14.4	3.1	18.0
Rift Valley	3,236	51.0	61.9	17.8	4.9	15.4
Western	1,481	55.1	62.7	18.9	2.3	16.2
Occupation						
Professionals & associates	753	23.4	59.9	4.7	10.6	24.7
Sales workers	1,656	40.1	69.7	8.0	6.4	15.9
Agricultural workers	4,561	59.8	52.8	26.8	1.5	18.9
Skilled workers	1,243	49.4	70.7	9.0	6.2	14.1
Unskilled workers	2,737	55.5	68.7	11.5	4.3	15.5
Other & no occp	2,030	48.0	65.1	10.8	4.1	20.1

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Annex Table 4: Share of Food Dietary Energy by Food Sources to Total Food Dietary Energy Consumption in Kenya

Categories and Groupings	Number of sampled households	Share of dietary energy purchased to total food consumption (%)	Share of dietary energy from own production to total food consumption (%)	Share of dietary energy eaten away from home to total food consumption (%)	Share of dietary energy from other sources to total food consumption (%)
Nationwide	12,980	57.7	13.4	3.3	25.6
Income level					
Quintile 1	2,596	54.8	13.5	0.8	30.9
Quintile 2	2,596	53.7	16.9	1.6	27.7
Quintile 3	2,596	58.2	16.7	2.1	23.0
Quintile 4	2,596	61.1	13.2	3.4	22.2
Quintile 5	2,596	60.0	6.7	8.2	25.2
Area					
Rural	8,382	52.4	18.3	1.8	27.5
Urban	4,598	68.4	3.6	6.4	21.6
HH Size					
1 to 2 people	2,576	58.2	8.1	8.6	25.0
3 to 4 people	3,452	59.9	12.3	3.7	24.0
5 to 6 people	3,473	56.6	14.3	2.6	26.5
7 or more	3,479	56.9	15.1	2.0	26.0
Gender of head of HH					
Male	9,108	57.6	13.7	3.8	24.9
Female	3,872	58.0	12.6	2.1	27.4
Age of head of HH					
Less than 35	4,084	63.1	9.3	4.8	22.8
Between 35 and 44	3,239	58.4	12.8	3.3	25.5
Between 45 and 59	3,363	56.3	15.1	2.9	25.6
Over 60	2,294	50.7	17.8	1.8	29.7
Economic activity of head of HH					
Employee	3,739	62.6	8.7	5.2	23.4
Self-employed	4,785	55.7	16.2	3.1	24.9
Other	1,732	50.3	19.7	1.9	28.2
No employment	2,724	59.9	10.1	2.2	27.9
Education of head of HH					
Pre-primary	3,324	56.6	13.0	1.5	28.9
Primary	5,634	57.8	15.4	2.9	23.9
Secondary and University	4,022	58.3	11.1	5.4	25.2
Region					
Nairobi	632	53.9	1.2	12.1	32.8
Central	1,460	62.8	15.5	2.7	19.0
Coast	1,234	61.9	8.9	3.3	25.8
Eastern	2,354	52.8	15.4	2.6	29.2
North Eastern	497	78.0	2.0	0.4	19.6
Nyanza	2,086	56.6	11.7	2.7	29.0
Rift Valley	3,236	57.0	16.7	3.8	22.5
Western	1,481	56.0	16.7	2.0	25.2
Occupation					
Professionals & associates	753	55.3	6.2	8.1	30.5
Sales workers	1,656	63.2	8.3	5.2	23.2
Agricultural workers	4,561	49.7	21.1	1.4	27.8
Skilled workers	1,243	64.9	9.0	5.4	20.7
Unskilled workers	2,737	63.5	10.6	3.6	22.3
Other & no occp	2,030	61.0	9.2	2.5	27.3

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Annex Table 5: FAO Indicators on Hunger Based on Total Consumption Expenditure and the First Quintile in Kenya

Categories and Groupings	Prevalence of food deprivation (%)	CV of dietary energy consumption FULL as FAO (%)	Average food consumption in dietary energy value (kcal/ person/day)	Minimum dietary energy requirement (kcal/person/ day)	Balanced dietary energy unit value of the first quintile (LC$/1000kcals)	Critical food poverty line of the 1st quintile	Total consumption expenditure (LC$/ person/day)	CV of total consumption expenditure -FULL (%)	Prevalence of Critical Food Poverty (%)	Income quintile having better balance nutritional diet
Nationwide	51.1	40.5	1,800	1,683	16.07	27.04	91.09	131.8	23.9	1
Area										
Rural	56.8	40.9	1,690	1,670	15.21	25.41	57.41	81.2	21.5	1
Urban	39.2	39.0	2,060	1,733	21.12	36.61	173.76	132.9	14.9	1
HH Size										
1 to 2 people	6.2	32.8	3,410	1,982	16.07	31.84	235.05	117.9	4.7	1
3 to 4 people	22.7	32.6	2,260	1,694	16.07	27.22	127.00	116.4	11.5	1
5 to 6 people	46.5	31.9	1,790	1,655	16.07	26.59	89.71	106.5	16.8	1
7 or more	74.5	34.4	1,410	1,659	16.07	26.65	59.03	95.3	27.9	1
Gender of head of HH										
Male	52.2	40.2	1,790	1,692	16.07	27.19	95.97	137.2	23.8	1
Female	48.5	41.9	1,820	1,657	16.07	26.62	77.84	112.3	23.1	1
Age of head of HH										
Less than 35	34.7	42.2	2,000	1,570	16.07	25.23	97.98	112.6	14.8	1
Between 35 and 44	52.9	39.0	1,710	1,641	16.07	26.37	94.36	129.4	21.5	1
Between 45 and 59	62.4	38.6	1,690	1,776	16.07	28.53	88.61	135.8	27.5	1
Over 60	57.4	44.2	1,840	1,724	16.07	27.70	80.76	148.7	32.6	1
Economic activity of head of HH										
Employee	44.8	39.2	1,910	1,693	16.07	27.20	124.76	119.6	12.6	1
Self-employed	47.8	38.7	1,840	1,679	16.07	26.97	90.54	131.1	23.8	1
Other	60.1	46.4	1,650	1,678	16.07	26.96	61.73	96.0	26.7	1
No employment	57.9	42.6	1,690	1,682	16.07	27.02	72.81	139.4	33.2	1
Education of head of HH										
Pre-primary	59.5	46.7	1,660	1,675	16.07	26.91	46.72	81.7	34.0	1
Primary	53.0	42.7	1,760	1,667	16.07	26.78	65.82	84.2	19.4	1
Secondary and University	41.1	37.3	1,990	1,716	16.07	27.57	173.21	135.1	9.8	1
Region										
Nairobi	14.7	31.5	2,530	1,745	23.35	40.73	482.17	129.2	2.3	1
Central	35.0	36.5	2,110	1,731	16.36	28.32	108.10	97.5	9.5	1
Coast	39.4	35.4	1,960	1,680	13.14	22.08	80.96	110.3	15.6	1
Eastern	27.0	36.7	2,240	1,695	11.40	19.33	71.11	98.9	12.2	1
North Eastern	66.3	42.6	1,460	1,595	15.22	24.27	44.77	79.8	30.1	5
Nyanza	68.1	42.7	1,490	1,666	23.93	39.87	84.52	90.6	28.0	1
Rift Valley	62.5	43.7	1,600	1,675	17.56	29.41	72.65	107.9	27.8	1
Western	69.7	48.0	1,440	1,639	22.61	37.06	65.30	87.7	35.5	1
Occupation										
Professionals & associates	33.4	35.1	2,150	1,751	16.07	28.13	328.36	134.3	2.8	1
Sales workers	38.8	37.3	2,020	1,708	16.07	27.44	135.70	101.8	7.0	1
Agricultural workers	57.3	42.0	1,690	1,676	16.07	26.94	56.21	78.7	23.8	1
Skilled workers	46.0	37.9	1,880	1,694	16.07	27.21	99.44	90.0	9.7	1
Unskilled workers	48.6	44.4	1,850	1,662	16.07	26.70	73.32	95.0	19.5	1
Other & no occp	59.5	44.8	1,660	1,683	16.07	27.05	77.53	150.8	33.6	1

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Annex Table 6: Inequality Measures (Dispersion Ratios) of Food Consumption, Total Consumption and Income in Kenya

Categories and Groupings	Dispersion ratio of food consumption in dietary energy (80/20)	Dispersion ratio of food consumption in monetary value (80/20)	Dispersion ratio of total consumption (80/20)	Dispersion ratio of income (80/20)
Nationwide	2.8	6.1	15.6	15.9
Area				
Rural	2.8	4.6	8.0	8.1
Urban	2.7	5.6	16.4	17.0
HH Size				
1 to 2 people	2.1	5.0	13.2	13.5
3 to 4 people	2.1	4.4	12.0	12.4
5 to 6 people	2.1	4.2	11.1	11.3
7 or more	2.4	4.2	9.7	9.8
Gender of head of HH				
Male	2.8	6.1	16.6	17.0
Female	2.9	6.0	12.9	13.0
Age of head of HH				
Less than 35	2.9	6.2	13.3	13.6
Between 35 and 44	2.6	5.9	16.0	16.5
Between 45 and 59	2.7	5.6	16.2	16.4
Over 60	3.1	5.9	15.2	15.4
Economic activity of head of HH				
Employee	2.7	5.9	15.1	15.3
Self-employed	2.7	5.4	14.5	14.9
Other	3.2	4.9	8.9	9.0
No employment	3.0	6.7	16.7	16.9
Education of head of HH				
Pre-primary	3.4	5.7	8.9	8.9
Primary	3.0	5.1	8.0	8.0
Secondary and University	2.5	5.6	16.5	17.1
Region				
Nairobi	1.9	5.5	23.2	24.7
Central	2.3	4.6	9.3	9.4
Coast	2.3	6.8	13.5	13.5
Eastern	2.5	5.7	11.2	11.4
North Eastern	2.9	4.8	8.1	8.1
Nyanza	3.0	4.5	8.9	9.0
Rift Valley	3.1	5.6	12.9	13.1
Western	3.4	4.4	8.6	8.6
Occupation				
Professionals & associates	2.1	5.6	20.6	21.8
Sales workers	2.4	5.2	11.5	11.6
Agricultural workers	2.8	4.5	7.4	7.4
Skilled workers	2.5	4.8	9.7	9.7
Unskilled workers	3.1	5.6	10.1	10.2
Other & no occp	3.2	7.8	20.2	20.7

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Annex Table 7: Inequality Measures (CV-Coefficient of Variation - Log Normal Assumption) of Food Consumption, Total Consumption and Income by Population Groupings in Kenya

Categories and Groupings	Number of sampled households	Average number of people in household	CV of food dietary energy DUE TO income (%)	CV of food consumption in monetary value DUE TO income (%)	CV of total consumption DUE TO income	CV of Income - FULL	CV of food dietary energy consumption FULL	CV of food consumption in monetary value FULL (%)	CV of food dietary energy consumption - FULL as defined by FAO (%)
Nationwide	12,980	5.0	35.2	68.5	131.8	134.2	61.9	83.2	40.5
Area									
Rural	8,382	5.5	35.7	54.0	81.2	81.4	62.5	67.5	40.9
Urban	4,598	4.1	33.5	65.2	132.9	136.5	58.1	83.3	39.0
HH Size									
1 to 2 people	2,576	1.5	26.0	60.8	117.9	120.4	54.5	73.8	32.8
3 to 4 people	3,452	3.6	25.8	55.5	116.4	119.5	49.3	64.7	32.6
5 to 6 people	3,473	5.5	24.9	50.4	106.5	107.8	50.5	62.1	31.9
7 or more	3,479	8.6	28.0	49.1	95.3	96.0	57.1	61.9	34.4
Gender of head of HH									
Male	9,108	5.2	34.9	68.8	137.2	139.9	61.8	84.3	40.2
Female	3,872	4.5	36.9	68.0	112.3	113.7	62.3	79.7	41.9
Age of head of HH									
Less than 35	4,084	3.8	37.1	70.4	112.6	115.5	59.3	81.9	42.2
Between 35 and 44	3,239	5.5	33.5	66.5	129.4	132.6	58.9	79.7	39.0
Between 45 and 59	3,363	6.1	33.1	63.7	135.8	137.5	62.3	81.2	38.6
Over 60	2,294	4.8	39.4	70.9	148.7	150.0	66.6	83.7	44.2
Economic activity of head of HH									
Employee	3,739	4.4	33.8	67.7	119.6	121.5	59.3	84.5	39.2
Self-employed	4,785	5.1	33.2	63.2	131.1	134.1	60.3	76.8	38.7
Other	1,732	5.6	41.9	57.7	96.0	98.0	68.9	75.3	46.4
No employment	2,724	5.1	37.6	74.0	139.4	141.0	62.3	86.9	42.6
Education of head of HH									
Pre-primary	3,324	5.4	42.2	62.8	81.7	81.7	65.2	73.1	46.7
Primary	5,634	5.0	37.8	59.0	84.2	84.2	63.3	72.6	42.7
Secondary and University	4,022	4.6	31.5	67.2	135.1	138.8	55.9	84.5	37.3
Region									
Nairobi	632	3.8	24.3	62.6	129.2	133.8	49.5	85.2	31.5
Central	1,460	4.1	30.5	57.8	92.5	92.5	51.1	72.2	36.5
Coast	1,234	5.3	29.2	69.6	110.3	110.2	52.5	89.8	35.4
Eastern	2,354	5.2	30.8	64.3	98.9	101.2	55.1	77.4	36.7
North Eastern	497	6.2	37.7	58.0	79.8	79.8	52.7	66.4	42.6
Nyanza	2,086	4.7	37.8	54.0	90.6	90.6	59.6	68.2	42.7
Rift Valley	3,236	5.2	38.9	62.7	107.9	109.2	64.4	77.0	43.7
Western	1,481	5.4	43.6	52.7	87.7	87.9	74.7	67.7	48.0
Occupation									
Professionals & associates	753	4.8	28.8	67.6	134.3	140.5	54.1	90.2	35.1
Sales workers	1,656	4.5	31.5	63.2	101.8	102.2	56.6	79.4	37.3
Agricultural workers	4,561	5.5	36.9	53.5	78.7	78.7	64.4	67.2	42.0
Skilled workers	1,243	4.5	32.1	57.5	90.0	90.4	55.9	70.9	37.9
Unskilled workers	2,737	4.7	39.7	63.0	95.0	95.7	62.4	74.3	44.4
Other & no occp	2,030	5.1	40.1	82.6	150.8	153.7	62.7	95.2	44.8

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Annex Table 7: Inequality Measures (GINI Coefficients-Free Distribution) of Food Consumption, Total Consumption and Income by Population Groupings in Kenya

Categories and Groupings	Number of sampled households	Average number of people in household	GINI of Food Dietary Energy Consumption DUE TO income (%)	GINI of Food Consumption in Monetary value) DUE TO income (%)	GINI of total consumption DUE TO income (%)	GINI of Income - FULL (%)	GINI of Food Dietary Energy Consumption – FULL (%)	GINI of Food Consumption in Monetary value) - (%)	GINI of Food Dietary Energy Consumption - FULL as defined by FAO (%)
Nationwide	12,980	5.0	19.6	33.7	51.4	51.8	31.4	38.2	21.7
Area									
Rural	8,382	5.5	19.6	28.4	38.9	39	32	33.3	21.9
Urban	4,598	4.1	18.7	32.8	52.8	53.5	28.9	38.1	21.0
HH Size									
1 to 2 people	2,576	1.5	14.8	32.1	50	50.6	28.4	36.6	17.9
3 to 4 people	3,452	3.6	14.5	28.6	48.9	49.6	25.8	32.5	17.8
5 to 6 people	3,473	5.5	14.2	27.2	47.2	47.5	26.7	32	17.4
7 or more	3,479	8.6	15.8	26.8	43.7	43.9	30.4	31.9	18.7
Gender of head of HH									
Male	9,108	5.2	19.4	33.8	52.6	53.1	31.4	38.4	21.6
Female	3,872	4.5	20.2	33.5	47	47.3	31.7	37.5	22.4
Age of head of HH									
Less than 35	4,084	3.8	20.4	34.3	47.6	48.1	30.1	37.7	22.5
Between 35 and 44	3,239	5.5	18.7	33.3	51.8	52.4	30.4	37.6	21.0
Between 45 and 59	3,363	6.1	18.4	31.7	52.2	52.5	31.8	37.2	20.8
Over 60	2,294	4.8	21.6	34	52.8	53	33.2	38	23.5
Economic activity of head of HH									
Employee	3,739	4.4	18.7	33.5	50	50.4	29.7	38.4	21.1
Self-employed	4,785	5.1	18.6	32.2	50.9	51.4	30.9	36.5	20.8
Other	1,732	5.6	22	29.2	41.5	41.9	34.6	35.2	24.5
No employment	2,724	5.1	20.7	35.4	52.3	52.6	31.7	39.2	22.7
Education of head of HH									
Pre-primary	3,324	5.4	22.7	32.1	39.8	39.8	32.9	35.8	24.6
Primary	5,634	5.0	20.2	29.6	38.7	38.7	31.9	34.3	22.8
Secondary and University	4,022	4.6	17.5	32.8	53	53.7	28.6	38.2	20.2
Region									
Nairobi	632	3.8	13.8	33.5	58	59.1	25.2	40.4	17.2
Central	1,460	4.1	17.1	30.5	43	43.1	26.2	35	19.7
Coast	1,234	5.3	15.9	34.3	48	48	26.2	40.1	19.2
Eastern	2,354	5.2	17.2	32.6	44.6	45.1	28.5	36.7	19.9
North Eastern	497	6.2	20	30.1	39.3	39.3	27.1	33.5	22.7
Nyanza	2,086	4.7	20.7	28.3	41.3	41.3	29.6	33.6	22.8
Rift Valley	3,236	5.2	21.5	31.5	46.2	46.4	32.3	36.1	23.3
Western	1,481	5.4	24.1	27.9	40.2	40.3	34.6	33.2	25.2
Occupation									
Professionals & associates	753	4.8	15.4	34.8	57	58.2	27.7	41.4	19.0
Sales workers	1,656	4.5	16.9	31	44.5	44.6	28.4	36.4	20.1
Agricultural workers	4,561	5.5	20.1	28	37.5	37.5	32.8	33.1	22.4
Skilled workers	1,243	4.5	17.6	30	41.7	41.8	28.4	34.3	20.4
Unskilled workers	2,737	4.7	21.3	31.9	42.4	42.5	31.4	35.7	23.6
Other & no occp	2,030	5.1	21.6	37.8	55	55.4	31.8	41.4	23.8

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Annex Table 9: Demand Elasticity with Respect to Income and Population Groupings in Kenya

Categories and Groupings	Income deciles	Income (KSh/person/day)	Demand elasticity of food dietary energy consumption (DEC) respect to income (%)	Demand elasticity of food consumption in monetary value (FMV) respect to income (%)	Demand elasticity of Engle ratio respect to income (%)
Income level					
Decile	1	16.65	0.8	-25.1	0.8
	2	29.02	0.5	1.9	0.8
	3	38.36	0.5	1.3	0.8
	4	48.37	0.4	1.0	0.7
	5	60.64	0.4	0.8	0.7
	6	75.96	0.4	0.7	0.7
	7	96.01	0.3	0.6	0.7
	8	127.28	0.3	0.5	0.7
	9	184.74	0.3	0.4	0.6
	10	571.00	0.2	0.3	0.3
Area					
Rural	1	14.34	1.3	8.3	0.8
	2	24.59	0.8	1.5	0.8
	3	31.82	0.6	1.1	0.8
	4	38.63	0.6	0.9	0.8
	5	46.30	0.5	0.8	0.8
	6	55.99	0.5	0.7	0.7
	7	68.19	0.4	0.6	0.7
	8	83.59	0.4	0.5	0.7
	9	111.49	0.4	0.5	0.7
	10	211.41	0.3	0.4	0.6
Urban	1	30.02	0.7	16.1	0.8
	2	52.90	0.5	1.6	0.8
	3	72.02	0.4	1.1	0.7
	4	91.60	0.4	0.8	0.7
	5	113.24	0.3	0.7	0.7
	6	139.01	0.3	0.6	0.7
	7	172.53	0.3	0.6	0.7
	8	227.73	0.3	0.5	0.6
	9	330.12	0.3	0.4	0.6
	10	1022.62	0.2	0.3	0.1

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Annex Table 10: Nutrient Density Per 1000 kcal in Kenya

Food commodity group	Average food dietary energy consumption (kcal/person/day)	Protein consumption (g/1000kcal)	Carbohydrates consumption (g/1000kcal)	Fats consumption (g/1000kcal)
Nationwide	1,800	29.5	169.6	23.6
Income level				
Quintile 1	1,080	28.2	176.1	21.5
Quintile 2	1,560	28.7	173.9	22.3
Quintile 3	1,910	29.4	171.6	22.9
Quintile 4	2,280	29.6	167.8	24.3
Quintile 5	3,020	31.1	159.9	26.4
Area				
Rural	1,690	29.3	172.8	22.5
Urban	2,060	29.8	163.1	25.7
HH Size				
1 to 2 people	3,410	30.4	164.5	24.5
3 to 4 people	2,260	29.8	167.3	24.3
5 to 6 people	1,790	29.5	169.6	23.7
7 or more	1,410	29.0	172.7	22.7
Gender of head of HH				
Male	1,790	29.5	169.4	23.5
Female	1,820	29.3	170.3	23.6
Age of head of HH				
Less than 35	2,000	29.6	166.6	24.7
Between 35 and 44	1,710	29.7	169.2	23.7
Between 45 and 59	1,690	29.4	170.5	23.2
Over 60	1,840	29.1	173.3	22.3
Economic activity of head of HH				
Employee	1,910	29.8	166.1	24.6
Self-employed	1,840	29.9	170.8	22.9
Other	1,650	29.0	171.2	23.3
No employment	1,690	28.5	170.8	23.6
Education of head of HH				
Pre-primary	1,660	28.7	174.0	22.4
Primary	1,760	29.5	171.8	22.6
Secondary and University	1,990	30.1	163.1	25.8
Region				
Nairobi	2,530	32.0	155.0	27.6
Central	2,110	29.7	167.9	23.7
Coast	1,960	28.2	171.2	22.7
Eastern	2,240	31.0	178.7	19.7
North Eastern	1,460	24.3	171.0	26.4
Nyanza	1,490	28.5	161.3	27.8
Rift Valley	1,600	30.0	166.1	24.6
Western	1,440	27.5	173.7	22.7
Occupation				
Professionals & associates	2,150	30.7	157.6	27.7
Sales workers	2,020	30.0	165.4	24.9
Agricultural workers	1,690	29.5	173.1	22.3
Skilled workers	1,880	29.9	165.9	24.6
Unskilled workers	1,850	29.2	171.8	22.7
Other & no occp	1,660	28.5	169.4	24.2

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The data do not consider wastage

Annex Table 11: Share of each Nutrient in Total Dietary Energy Consumption by Food Commodity Groups in Kenya

Food commodity group	Share of dietary energy consumption in Total Energy Consumption (%)	Share of protein consumption in Total Protein Consumption (%)	Share of carbohydrates Consumption in Total Carbohydrates Consumption (%)	Share of Fats Consumption in Total Fats Consumption (%)
CEREALS AND PRODUCTS	47.8	43.0	56.5	23.4
ROOTS AND TUBERS AND PRODUCTS	2.7	1.9	3.2	0.3
SUGARS AND SYRUPS AND PRODUCTS	9.2	0.0	14.4	0.1
PULSES	9.4	19.9	10.0	2.1
TREE NUTS	0.0	0.0	0.0	0.0
OIL CROPS	0.3	0.2	0.1	0.9
VEGETABLES AND PRODUCTS	4.1	5.0	4.9	0.7
FRUITS AND PRODUCTS	2.7	1.0	2.0	5.7
STIMULANTS	0.5	1.1	0.5	0.2
SPICES	0.1	0.1	0.1	0.1
ALCOHOLIC BEVERAGES	0.8	0.1	0.6	0.0
MEAT	1.7	9.5	0.1	2.4
EGGS	0.3	0.7	0.0	0.7
FISH AND FISH PRODUCTS	0.6	2.4	0.2	1.1
MILK AND CHEESE	7.1	11.1	3.1	17.5
OILS AND FATS (vegetable oils)	2.8	0.0	0.0	13.1
OILS AND FATS (animal fats)	5.8	0.0	0.0	27.5
NON ALCOHOLIC BEVERAGES	0.3	0.0	0.4	0.0
MISCELLANEOUS AND PREPARED FOOD	3.9	3.8	3.8	4.2
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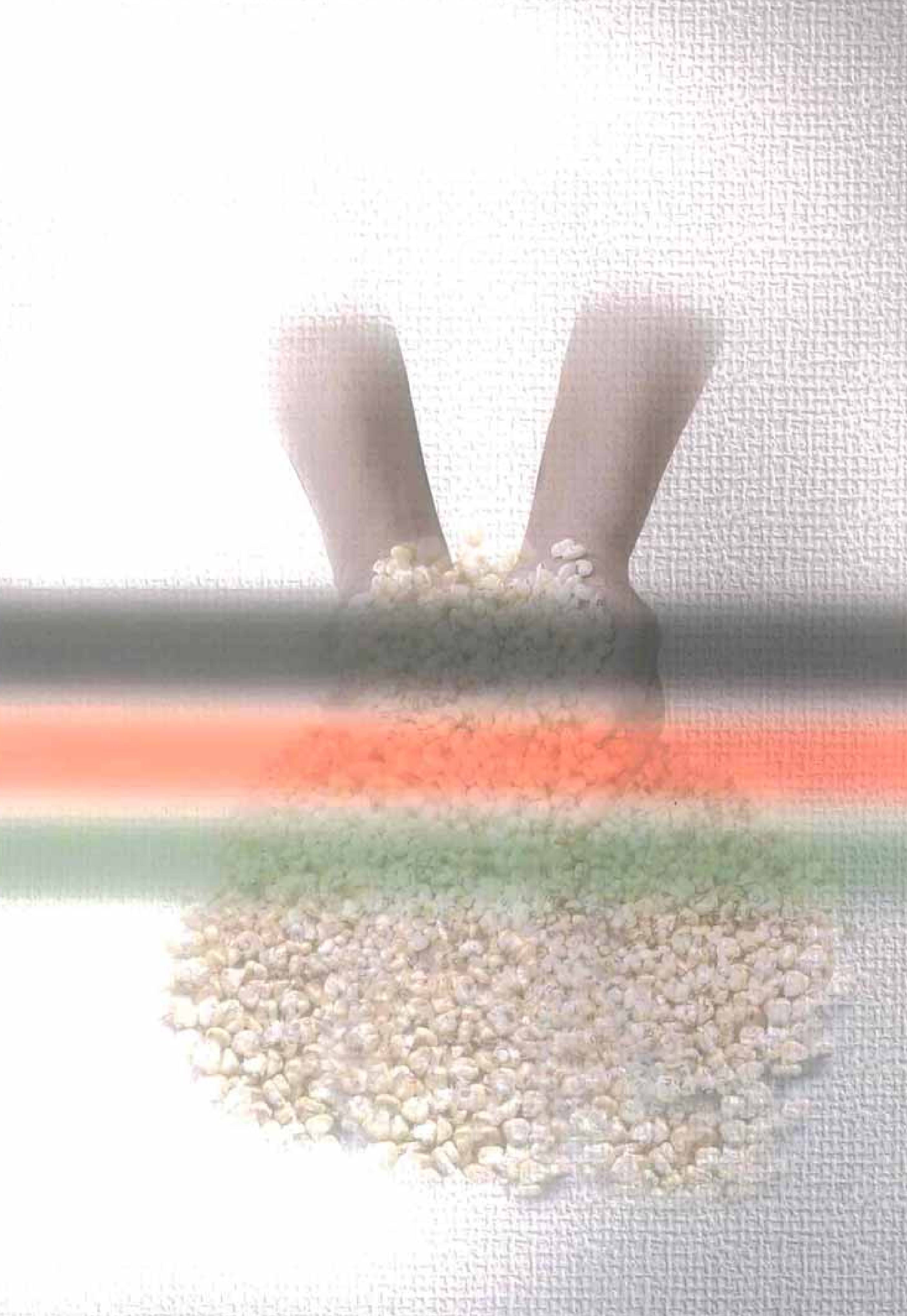
The data do not consider wastage



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